

VARIABILITY IN HARVEST INDEX OF GRAIN CROPS AND POTENTIAL SIGNIFICANCE FOR CARBON ACCOUNTING: EXAMPLES FROM AUSTRALIAN AGRICULTURE

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Abstract

For grain crops, harvest index (HI) is the ratio of harvested grain to total shoot dry matter, and this can be used as a measure of reproductive efficiency. The index can also be used to estimate crop carbon (C) balances by applying it to grain yield statistics to determine total shoot dry matter and then calculating crop residues as the difference between shoot C and grain C. Such an approach is widely used in C-accounting systems. Such a C-accounting practice is sensitive to changes in HI. In Australia, measured variations in HI are large enough to alter C balance calculations for some crops. Much of this variation results from the diverse range of climates and soils, which are a feature of the Australian cereal cropping region.

Factors that influence crop HI include the energy and protein content of seeds, long-term breeding achievements, and extreme (either hot or cold) temperatures during crop reproductive development. Crop husbandry can also influence HI, especially delayed sowing, which shortens the length of the vegetative phase and increases HI. For wheat, and perhaps some other C₃ cereals, excess nitrogen can enhance the allocation of photosynthate to structural carbon, which cannot be mobilized to grain later, resulting in a decrease in HI. Evidence for the balance between pre and postanthesis water use of field-grown crops having a significant influence on crop HI is equivocal.

A dataset containing more than 3000 estimates of HI in Australia has been assembled and used to summarize observed HI variations for each of the principal field crops grown in Australia. There remains a need for more reliable field HI data to be used in C-accounting systems and to aid the development of models to simulate likely regional and seasonal differences in HI for C-accounting purposes.

1. INTRODUCTION

The carbon balance of ecosystems has become a focus for research identifying sources and sinks for atmospheric CO₂ in an effort to develop strategies to mitigate anthropogenic emissions of greenhouse gases. Annual agricultural crop ecosystems are highly dynamic and provide different challenges compared to perennial forest and other perennial woody

vegetation systems, especially in Australia where the climate is more variable than otherwise apparently similar climates (Nicholls *et al.*, 1997). While growing crops initially represent a sink for atmospheric CO₂, the ultimate fate of assimilated carbon depends on the nature and use of crop products, the management of residues remaining after product harvest, the ability of a soil to accumulate and protect crop residue carbon against mineralization, and subsequent climatic conditions. Most approaches to C accounting in agriculture use measurements of harvested yield and estimates of the proportion of total plant carbon harvested to calculate the amount of carbon retained in the form of crop residues at any given location. Calculating the proportion of total plant carbon retained requires the use of defined values for the harvest index (HI), the ratio of carbon allocation to shoots and roots, and the carbon contents of harvested products and residues. This study reviews variations in harvest index for the major field crops grown in Australia. The amount of C in plant roots must also be included although we do not have space to consider this in the present analysis. Readers are referred to Bolinder *et al.* (1997), Gregory *et al.* (1995), Jackson *et al.* (1996), and Robinson (2004) for an introduction to shoot:root ratios and C accounting.

In agronomic terms, HI is calculated as the ratio of harvested product to total above-ground biological yield, which is an economic HI, and is usually calculated from unit area yield and dry matter data (machine-harvested yield/total plant dry weight at maturity). On a physiological basis, HI is a measure of the relative investment of plant resources in reproductive parts, and this is usually determined on a per plant basis and may include pod walls and other nonseed components. A full discussion on calculation of harvest indices can be found in Huhn (1990a,b) and the history and application of the index in Donald and Hamblin (1976), Evans (1993), and Sinclair (1998). None of these reviews were set within the context of agricultural carbon accounting, yet the HI data they reported have been relied upon to provide single HI values for individual crops that have been applied across regions, seasons (e.g., Adger and Subak, 1996; Bolinder *et al.*, 2007; Johnson *et al.*, 2006), and continents (e.g., Bondeau *et al.*, 2007; Lobell *et al.*, 2002). The magnitude of errors associated with such extrapolations is yet to be detailed, but based on Eq. (1), estimates of crop residues will be sensitive to changes in HI. This is illustrated in Fig. 1 where the amount of crop residue (Mg crop residue dry matter ha⁻¹) is calculated for a range of hypothetical HI and grain yield values (0.1–0.6 and 1–5 Mg grain dry matter ha⁻¹). Crop HI as expressed here refers to the aboveground portion of the crop, but for carbon-accounting purposes, it would also be necessary to account for the allocation of assimilated carbon to root dry matter. Including root carbon in the calculation effectively decreases HI and makes C accounting even more sensitive to changes in HI.

$$\text{Crop residues (Mgha}^{-1}\text{)} = \text{Grain yield (Mgha}^{-1}\text{)} \times \left(\frac{1 - \text{HI}}{\text{HI}} \right) \quad (1)$$

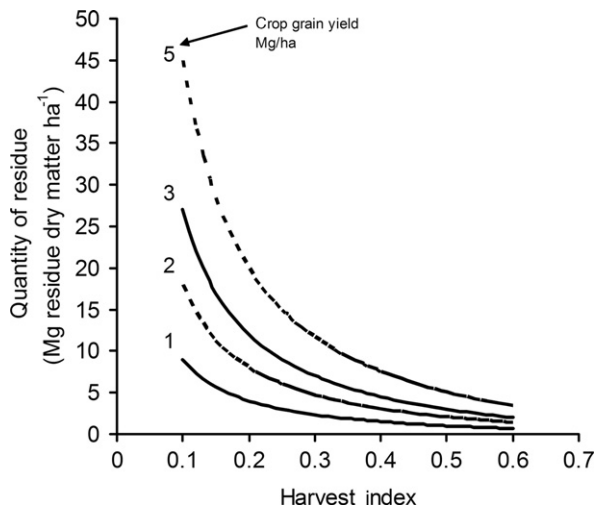


Figure 1 Change in the amount of crop residue retained at a site associated with variations in harvest index for grain yields of 1, 2, 3, and 5 Mg grain dry matter ha.

This review of harvest indices for Australian field crops represents the first step in a framework to develop protocols for assessing carbon balances in Australian cropping systems. In Australia, where winters are relatively mild, the majority of annual field crops (>90%) are autumn sown and harvested in early summer. Only a small fraction of the Australian crop area is devoted to summer crops such as sorghum, sugarcane, rice, sunflower, and maize. With total crop sowings exceeding 20 million ha annually and ranging over 40° of longitude and 25° of latitude, there is a considerable potential for the growth and development of field crops to vary with local seasonal conditions. Just over half of the Australian cropping area receives winter dominant rainfall, 40% has rainfall distributed more evenly throughout the year, and about 5% of the area is sown under a summer-dominant rainfall climate (Unkovich *et al.*, 2009). Across this cropping region, soils range from light sands to heavy clays, are variable in depth, can be uniform or exhibit strong textural contrasts, and may exhibit severe physical and chemical constraints within their subsoils (Adcock *et al.*, 2007; Dang *et al.*, 2006). Water holding capacity in the crop root zone varies from only 30 mm to more than 400 mm (McKenzie *et al.*, 2004), and potential seasonal cereal crop yields may thus vary from near 0 to more than 5 t/ha. Within this scenario, the application of single grain yield values and harvest indices for calculation of net C balances is unlikely to be reliable.

2. DETERMINANTS OF HARVEST INDEX

2.1. Energy investment in seeds, fruits, and storage organs

Comparisons among crop species reveal that the energy or carbon cost of producing seeds, the portion of the crop most often harvested, varies enormously. The quantity of seed that a given crop species can produce from a given amount of photosynthate (energy) is constrained by differences in plant biochemistry. In [Table 1](#), we assemble available data on seed composition and approximate energy (glucose) costs of synthesis (following [Penning de Vries *et al.*, 1974](#); [Sinclair and De Wit, 1975](#)). These calculations include the following assumptions:

- The principal energy costs are in synthesis of carbohydrates (CHO), proteins, and lipids; the equivalent glucose costs of which are protein from NH_3 —1.62 g glucose/g protein, protein from NO_3 —2.48 g glucose/g protein, carbohydrate—1.21 g/g, lipids—3.03 g/g, lignin—2.12 g/g, and organic acid—0.91 g/g (from [Penning de Vries *et al.*, 1983](#)).
- N for protein is acquired as nitrate for nonlegume crops (2.48 g glucose required/g N), and for legumes, the cost of symbiotic N fixation is the same as for ammonium assimilation (1.62 g glucose required/g N) ([Pate and Layzell, 1990](#))
- N requirement approximates mg protein N in seed/total glucose consumed (mg/g), and protein is 15% N

Carbon (energy) requirement, biomass productivity (PV, g seed/g glucose consumed), and N requirement can then be calculated according to [Eqs. \(2\)–\(4\)](#), respectively.

$$\text{mass glucose required (g)} = \sum (\% \text{CHO} \times 1.21) + (\% \text{protein} \times 2.48) + (\% \text{lipid} \times 3.03) \quad (2)$$

$$\text{PV} = \frac{\sum \% \text{CHO} + \text{protein} + \text{lipid} + \text{ash}}{\text{mass glucose required}} \quad (3)$$

$$\text{N requirement (mg/g)} = \frac{\text{protein} \times 0.15 \times 1000}{\text{mass glucose required}} \quad (4)$$

Our estimated productivity and N requirement data ([Table 1](#)) vary slightly from those of [Sinclair and De Wit \(1975\)](#), perhaps because we have used the conversion values of [Penning de Vries *et al.* \(1983\)](#) without rounding. The data should only be considered gross approximations for reproductive investment, as they do not include the cost of pods and other

Table 1 Carbohydrate (CHO), protein, lipid and ash content, biomass productivity (PV), and nitrogen demand of seeds of crop species

Crop	CHO (%)	Protein (%)	Lipid (%)	Ash (%)	PV (g/g)	N requirement (mg/g)	Refs.
Barley (<i>Hordeum vulgare</i>)	80	9	1	4	0.75	11	1
Rice (<i>Oryza sativa</i>)	88	8	2	2	0.75	9	1
Rye (<i>Secale cereale</i>)	82	14	2	2	0.72	15	1
Wheat (<i>Triticum esculentum</i>)	82	14	2	2	0.71	15	1
Oat (<i>Avena sativa</i>)	77	13	5	5	0.70	14	1
Mean for C3 cereals	82	12	2.4	3.0	0.73	13	
Sorghum (<i>Sorghum vulgare</i>)	82	12	4	2	0.70	13	1
Popcorn (<i>Zea mays praecox</i>)	80	13	5	2	0.69	14	1
Corn (<i>Zea mays</i>)	84	10	5	4	0.64	11	1
Mean for C4 cereals	82	12	4.7	2.7	0.68	13	
Bean, lima (<i>Phaseolus vulgaris</i>)	70	24	1.8	3.9	0.67	24	1, 3
Chickpea (<i>Cicer arietinum</i>)—kabuli	74	20	3.8	2.6	0.67	20	1, 3
Cowpea (<i>Vigna unguiculata</i>)	69	26	2	3	0.66	25	1
Faba bean (<i>Vicia faba</i>)	72	24	1.2	2.7	0.66	24	3
Field pea (<i>Pisum sativum</i>)	70	25	1.6	2.8	0.66	24	1, 3
Pigeon pea (<i>Cajanus cajan</i>)	69	25	2	4	0.66	25	1
Chickpea (<i>Cicer arietinum</i>)—desi	71	22	5	2.7	0.65	21	1, 3
Bean, mung (<i>Phaseolus aureus</i>)	66	26	4.5	3.8	0.64	25	1, 3
Lentil (<i>Lens culinaris</i>)	66	26	5	3.1	0.63	25	1, 3
Lupin (<i>Lupinus angustifolius</i>)	59	33	5.5	2.9	0.59	29	2, 3
Lupin (<i>Lupinus cosentinii</i>)	58	35	4	3	0.59	31	2
Vetch (<i>Vicia sativa</i>)	61	28	9	2.5	0.59	25	3
Lupin (<i>Lupinus albus</i>)	51	37	9.2	3.2	0.56	31	2, 3
Lupin (<i>Lupinus luteus</i>)	49	42	5	4	0.56	35	2
Soybean (<i>Glycine max</i>)	38	38	20	4	0.50	28	1

Table 1 (continued)

Crop	CHO (%)	Protein (%)	Lipid (%)	Ash (%)	PV (g/g)	N requirement (mg/g)	Refs.
Peanut (<i>Arachis hypogaea</i>)	25	27	45	3	0.43	17	1
Mean for legumes	60	29	7.8	32	0.61	26	
Cotton (<i>Gossypium hirsutum</i>)	47	25	25	3	0.52	19	1
Safflower (<i>Carthamus tinctorius</i>)	50	14	33	3	0.52	11	1
Sunflower (<i>Helianthus annuus</i>)	48	20	29	3	0.51	15	1
Flax (<i>Linum usitatissimum</i>)	32	26	38	4	0.46	18	1
Hemp (<i>Cannabis sativa</i>)	27	29	41	3	0.44	19	1
Canola (<i>Brassica napus</i>)	25	23	48	4	0.43	15	1
Sesame (<i>Sesamum indicum</i>)	19	20	54	7	0.42	13	1

CHO data for [Pettersen and Mackintosh, 1994](#) (Ref. 2 data) is approximated as $(100 - (\text{protein} + \text{lipid} + \text{ash}))$.

We have recalculated the PV and N requirement of [Sinclair and De Wit \(1975\)](#) (Ref. 1) and [Perry et al. \(1986\)](#) (Ref. 2) and added data from [Pettersen and Mackintosh \(1994\)](#) (Ref. 3).

reproductive structures, but only seeds. Some protein content conversions also suffer from estimation, and some seed components are left out (lignin, organic acids), although these come with very little C cost ([Penning de Vries et al., 1974](#)). Nevertheless the rankings among crop species are valid.

On average, C3 cereal crops turn out to be the most efficient in their conversion of photosynthate to grain with a biomass productivity (PV) averaging 0.73, due to the low lipid and protein contents of their seed. Barley is highlighted as being the most efficient. [Flood and Martin \(2001\)](#) noted a negative relationship between grain protein content and HI in wheat. Legumes tend to have a higher carbon cost for seed production, averaging 0.61 g seed dry matter/g glucose invested. Peanut stands out as having a very high cost of seed production (0.43 g/g) due to high seed oil content (45%). This is similar to a range of other high oil species such as cotton, canola, and sunflower.

On a biochemical basis one would thus expect cereals to have higher harvest indices than grain legumes, which in turn would be higher than the other species with an oil-rich seed ([Table 1](#)). Compared to noncereal crops,

the plasticity in grain protein concentration of wheat (Flood and Martin, 2001), and other cereals, may reduce variations in harvest index. In contrast, Lupin (*Lupinus angustifolius*) seed, for example, tends to have a much more stable protein content (Duthion and Pigeaire, 1993) and grain N concentration may dictate grain yield more so than for cereal crops, although N supply such in N₂ fixing legumes should be nonlimiting.

Manufacture of seeds and other reproductive structures comes at a significant energy cost, both in terms of storage and biosynthesis, especially where seeds contain lipids and proteins, and thus seed biology impinges directly on the amount of seed that can be produced for a given amount of photosynthate.

2.2. Breeding

The HI of crop plants has increased over time due to breeding for higher yield (see Calderini *et al.*, 1999; Perry and D'Antuono, 1989; Whitehead *et al.*, 2000), and more recently, specifically for HI (Sinclair, 1998). Based on the data presented in (Fig. 2) and Turner (1997), we calculate that the rate of HI increase due to cereal breeding in Australia is about 0.015 per decade, compared to about 0.02 achieved in the United Kingdom (Turner, 1997). Shorter-statured, modern crop cultivars have higher harvest indices than their taller forebears, although total dry matter production is most often very similar (Evans, 1993). A recurring theme in HI gain has also been a shortening of the vegetative phase of crop plants, providing a proportionately

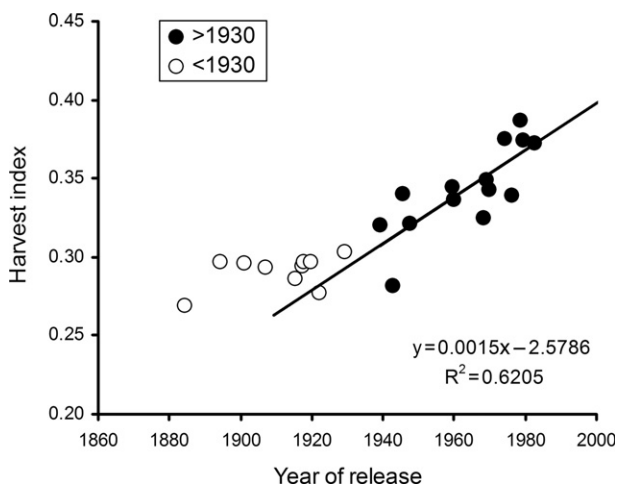


Figure 2 Changes in harvest index of wheat for Australian wheat varieties 1884–1983, all grown between 1979 and 1981 at a range of sites in Western Australia (redrawn from data of Perry and D'Antuono, 1989).

longer grain-filling period (Lopez Pereira *et al.*, 2000; Siddique *et al.*, 1989). A similar effect can often be achieved by delayed sowing (e.g., Batten and Khan, 1987b; Gomez-Macpherson and Richards, 1995). These increases in HI of crops over time may be an unrecognized source of error where C accounting is conducted over extended historical time periods (e.g., Adger and Subak, 1996) in which significant shifts in HI have occurred and crop residue returns are calculated from measured grain yields using a constant harvest index.

2.3. Determinacy

Most cereal crop cultivars are *determinate*, flowering within a very small window of time and then entering a grain-filling period such that vegetative and reproductive growth are separated in time. Determinate crops generally have higher harvest indices as most crop resources are diverted to grain production once flowering has commenced. In contrast, for *indeterminate* crops, such as most legumes and oilseeds, concurrent vegetative and reproductive growth is maintained, resulting in competition for resources between vegetative and reproductive sinks (Constable and Gleeson, 1977; Pate and Armstrong, 1996), and flowering occurs over an extended period during which flowers and pods may be subjected to a range of climatic stresses. Harvest index could thus be more variable in indeterminate than determinate crops. In cases where both determinate and indeterminate cultivars are available (e.g., faba bean, soybean), the determinate varieties tend to have higher harvest indices (De Costa *et al.*, 1997; Erskine *et al.*, 1988). While determinate crops are also thought to have less variable harvest indices, we cannot find sufficient data to reliably assess this. While flowering over an extended period in indeterminate crops means that developing flowers and pods are subject to a broader range of climatic conditions, some of which may cause abortion of reproductive parts, it also means that aborted flowers or seeds may be compensated for by later formed flowers.

2.4. Effects of pests and diseases

Crop harvest indices can be reduced by pests and diseases through impacts on flowers or seed and pod development, or indirectly through destruction of grain. Diseases and pests can reduce kernel number or size, for example, Hoffman and Kolb (1998) showed that HI in wheat was correlated with the incidence of bean yellow mosaic virus, and McKirdy *et al.* (2002) showed that this virus caused shriveled grain in wheat, but less so in oat. In Australia pea weevil (*Bruchus pisorum*) and other insect pests can inflict significant yield losses on grain legume crops (Hardie *et al.*, 1995), significantly reducing economic yield and harvest index.

2.5. Temperature extremes during flowering and grain fill

For most plants reproductive development is sensitive to a wide range of environmental impacts, with many grain crops having a preferred temperature range for flower and seed development. Large variations in temperature outside of this range may cause problems with flower or seed growth. Damage can be caused by freezing injury or due to temperatures just above freezing point. A definition of absolute values of these is problematic at the field level as there will be some differences among crop species, cultivars and age, and interactions with local landscape factors. For chickpea, Croser *et al.* (2003) defined freezing temperatures as $< -1.5^{\circ}\text{C}$ and “chilling temperatures” between -1.5 and 15°C . For other crops, the freezing temperatures are likely to be similar, but the range of chilling sensitivity will vary considerably. Regardless of crop species, in Australia, chilling temperature effects will be much more common than freezing temperatures. Where seed or flower development is constrained due to specific effects on reproductive apparatus, harvest index is likely to be significantly reduced, especially for determinate crops. The simplest example of a temperature extreme is frost, with freezing injury causing flower or seed abortion. In Australia this is mostly managed by combining sowing time with cultivar such that crop anthesis occurs after the last heavy frost (Boer *et al.*, 1993; Hodgson, 1978). This approach is mostly successful, but severe frosts after anthesis still occur on occasion and cause reductions in HI and grain quality (e.g., Allen *et al.*, 2001). With a detailed knowledge of crop sowing date and cultivar, the likelihood of frost damage can be estimated if combined with local historical weather measurements (Robertson *et al.*, 2001), but the actual impact on yield or HI cannot be modeled as the actual damage to grain caused by frosts cannot be accurately predicted within a crop, region, or field. In Australia frost events may not have a major impact on HI due to flexibility in sowing dates, the range of crop cultivars available, the relatively low incidence of frost occurrence, and its localized nature, except in exceptional ($\geq 1/20$) years.

Other temperature stresses on reproductive performance of crops also affect HI. Working with canola in low rainfall WA, Si and Walton (2004) found that high postanthesis rainfall and low postanthesis temperature correlated with higher yield. Pod temperatures of $34\text{--}38^{\circ}\text{C}$ are known to cause seed abortion in lupin (Reader *et al.*, 1997) and occur in about 1 year in three in the main lupin-growing region of WA, but impacts on crop HI were not measured. While one might expect pod abortion to result in a lower HI, in the case of lupin, an indeterminate crop, later formed pods can compensate to some extent (Seymour, 1996), but at the field level this is likely to be the exception rather than the rule (Dracup, 1994). High postanthesis temperatures can also reduce grain yield and quality in wheat, but with substantial differences in sensitivity among cultivars (Stone and Nicholas, 1995).

2.6. Pre:Postanthesis water use

Total water use (evapotranspiration) may not always be the principal limitation to grain yield for rainfed grain crops in Australia (Cornish and Murray, 1989; Passioura, 2006), but the seasonal distribution of rainfall and plant available soil water relative to current crop demand and crop development is often thought to be critical (Angus and van Herwaarden, 2001). In Australia this has been most extensively studied in wheat (e.g., van Herwaarden *et al.*, 1998a), but the effect of timing of rainfall has also been examined in field experiments with barley (Goyne *et al.*, 1993), lupin (Palta *et al.*, 2003), and canola (e.g., Wright *et al.*, 1992). In this last study, canola grown in northern NSW was either rainfed, irrigated during pod fill, or grown only on stored soil water after sowing (using rain out shelters). Harvest index decreased in the order irrigated > rainfed > rain excluded. Thus having more water available to canola plants postanthesis increased harvest index. This relationship was shown quite eloquently for wheat in the glasshouse experiments of Passioura (1977), as shown in Fig. 3. In both these experiments HI was found to be independent of total water use. Richards and Townley-Smith (1987) found a similar relationship in later glasshouse experiments. The curvilinear relationship they observed (Fig. 4) was attributed to a variable contribution of stored C from preanthesis photosynthesis to grain yield.

It is not clear how applicable these essentially glasshouse-derived relationships between pre/postanthesis water use and HI are to actual field conditions. Field data to assess this are scarce. For rainfed field crops it is difficult to manage water supply to match current crop demand. The balance between pre- and postanthesis growth will be a function of available

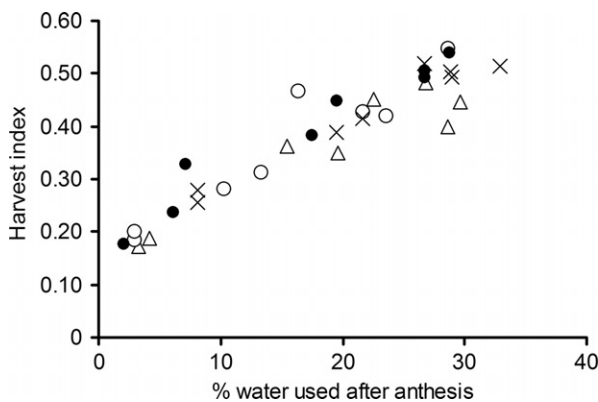


Figure 3 Harvest index of pot cultured wheat as a function of the percentage of total water supply used after anthesis (replotted from Passioura, 1977 with permission). Different symbols are for different cultivars.

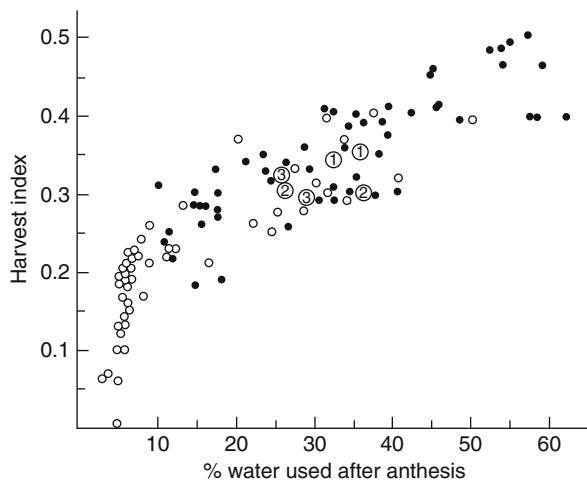


Figure 4 Relation between HI and the percentage of total water used after anthesis for winter (○) or spring-summer (●) sown wheat in the glasshouse (reproduced from Richards and Townley-Smith, 1987 with permission).

soil water (rainfall $\times$ soil type $\times$ soil evaporation) and the rate it is used by the crop. Preanthesis transpiration can be increased by promoting more rapid development of crop leaf area. This can be done with the application of fertilizers, especially N, and with crop rotations which reduce root diseases and promote faster root growth. Together these could lead to faster leaf area development and earlier exhaustion of soil water.

Field experiments in NSW employing irrigation and rain out shelters (Kirkegaard *et al.*, 2007) examined seasonal water use and yield of wheat. In one experiment additional postanthesis water use of ca 10 mm increased grain yield but had no impact on harvest index, while in a second experiment where a severe water stress was imposed, extraction of an additional 30 mm of water increased harvest index from 0.31 to 0.38.

The only substantial field datasets on pre- and postanthesis water use and crop HI that we have been able to find for given sites in Australia are assembled in Fig. 5. Here it can be seen that there is little evidence for a relationship between pre:postanthesis water use and HI of these field-grown wheat crops. Averaging across site $\times$ year may mask variations in fractional postanthesis water use and HI due to crop sequence (e.g., Lenssen *et al.*, 2007; O'Connell *et al.*, 2002); however, we found that if data were plotted without averaging site $\times$ year, the correlation was even weaker than that seen in Fig. 5. Wahbi and Gregory (1989) presented a negative relationship between the percentage of total water use (20–35%) after anthesis and HI of barley grown in Syria and concluded that water use during grain filling was a poor indicator of harvest index.

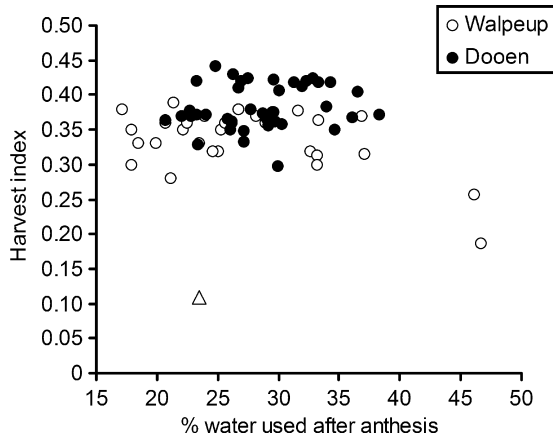


Figure 5 Scatter plot of percentage water used after anthesis and harvest index of wheat at Dooen (Cantero-Martinez *et al.*, 1995) and Walpeup (from O'Connell *et al.*, 2002; O'Leary and Connor, 1997) in Victoria. The triangle represents an outlier in the Walpeup dataset. Excluding this outlier there was a weak negative correlation between the fraction of water used after anthesis and wheat harvest index ($R^2 = 0.39$, $HI = -0.1905 \times \text{fraction water use after anthesis} + 0.3217$).

Reasons for a weaker relationship between pre:postanthesis water use and harvest index of wheat in field versus glasshouse studies could be that the range in % of water used after anthesis for field-grown crops at a given site may be less than that engineered in glasshouse experiments, or in field experiments employing irrigation (e.g., Goyne *et al.*, 1993; Palta *et al.*, 2003; Wright *et al.*, 1992). For example, for wheat over a 10-year period (1981–1991) at Dooen in Victoria, postanthesis water use of wheat varied from 18% to 31% of total crop water use, but with 70% of values being between 25% and 30%. At Walpeup, a drier environment in north-west Victoria, postanthesis water use of wheat similarly ranged from 18% to 46% of total water use. This is probably the least sensitive of the range in fractional water use in the data of Richards and Townley-Smith (1987) shown in Fig. 4.

The plasticity of the wheat plant may also contribute to reduced variation in harvest index at the field level, with the number of ears m^{-2} , the number of spikelets per ear, number of grains per spikelet, and grain size all being variable and allowing for considerable compensation among yield components, especially since these components are formed sequentially. Other factors, such as shoot:root development (Shepherd *et al.*, 1987) and mobilization of preanthesis assimilates to grain, can operate to stabilize harvest index at the field level (Palta and Fillery, 1995; van Herwaarden *et al.*, 1998a).

Relationships between pre and postanthesis water use and harvest index have also been reported for indeterminate plants. In glasshouse-grown lentil (Shrestha *et al.*, 2006) harvest index ranged from 0.36 to 0.45 when plants were water stressed during the reproductive phase, compared to 0.53–0.59

for well-watered lentil. Reductions in seed yield due to water stress during reproductive development resulted from reduced flower number and from seed abortion, but not seed growth rate or seed size. In this case, postanthesis water use was proportionately lower in the droughted treatment. Water stress during chickpea pod development resulted in pod drop and lower harvest indices (Thomas and Fukai, 1995). For lupin crops in Western Australia, pod fill does not commence until the majority of soil water has been used, restricting pod fill and harvest index (Dracup *et al.*, 1998a). Efforts to reduce competition for assimilates between concurrent vegetative and seed pod growth in lupin, by breeding lines with reduced levels of branching, have contributed to increased yield (Hamblin *et al.*, 1986), most probably through earlier flowering, but with only a slight increase in harvest index (Dracup *et al.*, 1998a). In glasshouse-grown lupin, transient water deficits reduced dry matter production by 28% and seed yield by 33%, but harvest index was only reduced by 9% (Palta and Plaut, 1999). Identifying such small differences in HI due to fractional water use at the field level is problematic. Significant positive HI responses to late season watering have been demonstrated in irrigation experiments for crops such as lupin (Dracup *et al.*, 1998a), chickpea (Leport *et al.*, 1999), and canola (Wright *et al.*, 1992) in Australia, but substantial field datasets for rainfed crops to clearly demonstrate direct effects of differences in fractional water use postanthesis are not readily apparent.

Across most of the Australian cropping zone crop water deficits during pod and grain fill are the norm (Nix, 1975), while alleviating this can potentially increase grain yields and crop harvest index, increased parallel vegetative growth may act to moderate significant positive impacts of this in indeterminate broadleaved crops. The ability of broadleaved crops to translocate C from vegetative parts to pods or seeds varies among species and among cultivars within species. Leport *et al.* (1999) highlighted that this may be a way in which HI of chickpea might be improved. Lupin on the other hand appears to have a limited capacity to mobilize earlier fixed C and is more reliant on current photosynthesis to fill seeds (Pate *et al.*, 1980).

The variable carbon cost of seed production for different species and varieties may mask the relationship between seasonal water use and HI. Although Sadras and Connor (1991) were able to take this into account and replicate the relationship seen in Fig. 4 in response to fractional water use for the data of Richards and Townley-Smith (1987), and for container-grown sunflower in the field, recalculating the harvest index production value (HI_{pv}) for the data of Fig. 5 did not strengthen the relationship between the fraction of water used after anthesis and HI.

Under field conditions where water supply is not controlled, other factors may counterbalance any effects of fractional seasonal water use on HI as observed in glasshouse or controlled environment studies, and field evidence for fractional seasonal crop water use impacting on crop HI is weak. A similar conclusion was reached by Turner (1997).

2.7. Water use, soil mineral nitrogen, and harvest index of wheat

The field studies of [van Herwaarden *et al.* \(1998a\)](#) provide some evidence for there being a relationship between fractional water use after anthesis and harvest index; however, this dataset is confounded by interactions with N fertilizer applications. Indeed N fertilizer rates may have an influence on HI that does not operate through seasonal water use patterns. The work of [van Herwaarden *et al.* \(1998a,b\)](#) indicates that high soil N fertility might result in a decrease in the deposition water-soluble carbohydrates in wheat stems, tillers, and leaves and an increase in structural carbon, leading to less mobilization of vegetative C to grain and a lowering of harvest index. Examining wheat crops in SE Australia, [Angus and van Herwaarden \(2001\)](#) concluded that vigorous preanthesis growth can increase plant transpiration, at the expense of soil evaporation, and lead to a higher C fixation. However, structural carbohydrate laid down earlier in the growing season is not readily mobilized to the grain. In irrigation experiments with wheat receiving 44 kg N/ha ([Kirkegaard *et al.*, \(2007\)](#)), a greater fraction of stem water-soluble carbohydrates were mobilized to grain in more water-stressed plants, but this did not result in increased harvest index. In a second experiment, there were no significant differences in water-soluble carbohydrates between irrigation treatments.

Replotting the HI dataset of [Cantero-Martinez *et al.* \(1995\)](#) on the basis of available N ([Fig. 6](#)) shows a weak negative relationship between available N at sowing and the harvest index of wheat, whereas no relationship was apparent for fractional water use after anthesis and crop harvest index in this same dataset. A similar relationship is indicated in the data of [Adcock \(2006\)](#) from the Eyre Peninsula SA. Here harvest index of wheat in 2 years was

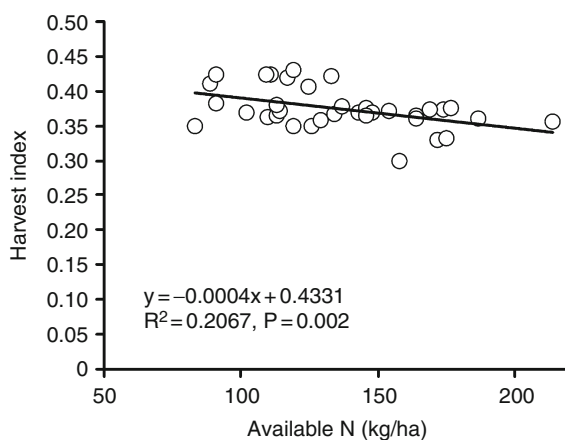


Figure 6 Correlation between soil nitrate-N at sowing (0–1 m) and harvest index of wheat at Doon Victoria (1981–1990). Data from [Cantero-Martinez *et al.* \(1995\)](#).

lower after legumes (vetch or medic) than after nonlegumes (canola, barley, or wheat), although there were minimal differences in the fraction of water used after anthesis between rotations within a given year. The timing of seasonal water use was governed by absolute water availability (rainfall) and not through deferral of water use within the crop-growing season. In this study, soil mineral N was higher when the preceding crop was a legume (133–225 kg N/ha) than cereal or canola (81–149 kg N/ha), and this may have contributed to the differences in HI rather than seasonal water use patterns by the crops.

The importance of nitrogen availability in Western Australia is yet to be determined, but the generally lower soil fertility, lower N fertilizer applications resulting from a greater uncertainty of responses to N fertilizers due to climate variability, and cultivars with greater concentrations of water-soluble carbohydrates ([van Herwaarden and Richards, 2002](#)), may reduce the likelihood of poor mobilization of stored carbohydrates to grain in the western wheatbelt. In a pot study on barley, [Fathi *et al.* \(1997\)](#) found that reductions in grain yield and increases in grain protein due to postanthesis water stress were greater when plants were grown under high N supply than when the supply of N was limited. This result is in concordance with the aforementioned studies on wheat. A theoretical framework relating seasonal water use and remobilization of stored C was presented by [Fischer \(1979\)](#).

2.8. Summary of factors influencing HI

In Australia, environment rather than genotype is seen as the major determinant of harvest index for most field crops for a given site $\times$ year $\times$ genotype ([Turner *et al.*, 1999](#)). Frost and cold temperatures at flowering, or high temperatures during flowering or seed development can contribute to substantial variation in HI. The balance between pre- and postanthesis water use is not yet clearly implicated in variation in HI of field crops, but in some regions vigorous growth early due to high soil N status, perhaps in conjunction with available water, reduces HI of wheat, and possibly barley. Field data on seasonal water use and HI for other crops are scant.

3. A DATABASE OF CROP DRY MATTER, GRAIN YIELDS, AND HARVEST INDICES

We have collated data on grain yield, crop dry matter, or harvest index for a range of crops grown in Australia. Data were from readily available literature and from generous contributions from colleagues. We restricted our data search to ≥ 1980 as changes in crop growth and yield over time due to agronomy and breeding (yield) may otherwise confound some

interpretations. Data were included from small plot experiments, from farmer-managed paddocks, and in the case of HI only, also from glasshouse experiments. At the time of writing >365 data sources have been used to accumulate 20,100 cereal crop entries, 4500 legume crop entries, and 5500 entries for canola crops. The assembled data are available as accessory material on the publisher's Web site (URL). It should be interpreted with some caution since it may not be a representative sample of grain crops across Australia, but is a sample of primarily research data from an assemblage of experiments and sites which were not distributed equally across the grain-growing regions in time or space. Differences between experimental yields and farmer best practice yields are not known. In cases where commercial crops have had the same inputs as experimental plots, yields have been equivalent (Evans, 1993). However, there is a tendency for experimental plots to be located on better land within a field, and for inputs to be higher than for commercial crops, thus perhaps skewing research data to achievable but more economically risky yields.

3.1. Problems with interpretation of agronomic harvest indices

Grain/produce yield data are usually reported on an "as harvested" basis. Typically this means that they are not reported on a dry weight basis, but at some nominal moisture content, conforming to the receival quality assurance standards dictated by wholesalers or processors. For some crops at physiological maturity the moisture content may be as high as 30% (e.g., maize, Table 2), but this may be left to dry out, or managed to obtain a lower moisture content before and/or after harvest. Many reports of grain yield do not indicate moisture content, even in the research literature. In our database we have not specifically corrected for moisture content and this would introduce some error.

In addition to moisture content, variations in calculated HI values can occur depending on the timing of sampling or the extent of leaf drop and crop attrition during senescence. For crops which drop leaves prior to maturity, HI should be calculated from peak biomass dry matter, including fallen leaves, not just from standing dry matter at maturity. For crops which shed leaves, HI calculated in this latter way is referred to as "apparent harvest index" (Schapaugh and Wilcox, 1980), and these are usually higher than actual HI (e.g., apparent harvest index of ≥ 0.75 for soybean Mayer *et al.*, 1991). Some of the harvest indices in our database might be *apparent* rather than actual HI *per se*, and comparisons among crop species should be carefully considered. We tried to exclude any data which did not include fallen leaves although this was rarely explicitly stated within the original data sources used.

Table 2 Typical moisture content for harvested material, and harvest losses due to shattering/harvest inefficiencies

Crop	Typical moisture content of harvested product (%)	Source	Typical harvest losses (%)	Source
Wheat	11	Receiveal standard ^a		
Barley	11	Receiveal standard		
Canola	7	Receiveal standard		
Lupin	> 12	Dracup <i>et al.</i> (1998a,b)	20	Snowball (1986)
Grain sorghum	≥ 13.5	Receiveal standard		
Oats for grain	11	Receiveal standard		
Field pea	≥ 14	Receiveal standard	5–24	Cassells and Armstrong (1998)
Triticale	11	Receiveal standard		
Chick pea	≥ 14	Receiveal standard	10–20	O'Mara (undated)
Faba bean	≥ 14	Receiveal standard		
Rice	21	Receiveal standard		
Lentil	≥ 14	Receiveal standard		
Mung bean	9	Petterson and Mackintosh (1994)		
Sunflower	9	Receiveal standard		
Maize	13.5	Spenceley (2005)	22	Robertson <i>et al.</i> (2003)
Soybean	15	Rose <i>et al.</i> (2005)		

^a Receiveal standards are maximum values for Australian grain delivered to silo, average values are likely to be 1–2% lower.

The influence that harvesting method (hand or machine harvested) and inclusion of fallen leaves can have on calculated harvest index was indicated for lupin in Western Australia (Table 3). The data showed that HI varied from 0.16 to 0.29 depending on the rigor of sampling, and crop shoot dry matter attrition between peak dry matter (DM) and harvesting, leading to an apparent increase in harvest index. Similarly, Dracup (1994) estimated harvest index of a lupin crop to be 0.19 at the time of harvest, but 0.15 if fallen leaves were also taken into account. For pot-grown chickpea, Hooda *et al.* (1986) calculated harvest indices of 0.25 if fallen leaves were excluded

Table 3 Harvest index for a lupin crop calculated from various sampling strategies (from [Unkovich, 1991](#))

HI formula used	HI
Hand-harvested seed/standing biomass	0.29
Mechanically harvested seed/standing biomass	0.21
Hand-harvested seed/(standing biomass + fallen leaves)	0.22
Mechanically harvested seed/(standing biomass + fallen leaves)	0.16
Hand-harvested seed/peak standing biomass	0.20

Table 4 Harvest index of chickpea (*Cicer arietinum*), calculated for C, N, and dry matter, and including different plant fractions (from [Hooda et al., 1986](#))

	Carbon	Nitrogen	Dry matter
Including above-ground biomass only			
Excluding fallen leaves	0.26	0.50	0.25
Including fallen leaves	0.20	0.38	0.18
Including roots			
Excluding fallen leaves	0.20	0.37	0.16
Including fallen leaves	0.16	0.31	0.13

and 0.18 if they were included ([Table 4](#)). [Saxena et al. \(1983\)](#) indicated that excluding fallen leaves (pinnae) from HI calculations reduced HI in chickpea by 10%. In experiments with canola ([Hocking et al., 1997](#)) HI was estimated to average 0.28, or 0.25 if shed leaves were included. Even where fallen leaves were carefully collected with sequential samplings, total dry matter of lupin crops declined between a peak around midgrain fill and maturity ([Greenwood et al., 1975](#); [Unkovich et al., 1994](#)). [Prince et al. \(2001\)](#) used the relationship between apparent and actual HI in soybean of [Schapaugh and Wilcox \(1980\)](#) for estimating crop DM from regional yield data (Actual HI = $-9.76 + 0.96 \times \text{apparent HI}$). Whether such relationships have general applicability is not known and we could not find sufficient data to derive such relationships for field-grown crops in Australia.

Reported HI values may also be influenced by the proportion of yield produced that is harvestable. Lodging due to weak stems, storm events, insect damage, or other processes can reduce the proportion of grain produced that is harvested and thus induce error in HI estimates or estimates of crop residues where average HI values are used. [McDonald \(2006\)](#) reported hand-harvested wheat yields in a commercial crop to be double that of the machine-harvested yields. On the upper Eyre Peninsula SA, harvest losses as high as 17.5% for field pea and 58% for lupin have been reported ([Egan et al., 1994](#)). For short-statured crops such as lentil and

chickpea, many of the lower pods may remain unharvested, especially on uneven ground. [Schwenke *et al.* \(1998\)](#) compared farmer yields with measured (quadrat) yields in commercial crops over 2 years in northern NSW. On average, farmers reaped only 48% of the hand-harvested chickpea yields and 65% of faba bean yield. Problems with low harvestability and high shattering losses are probably a lesser issue in research plots which are more intensively managed than commercial crops. Nevertheless it illustrates that in some instances, plot yields, harvested either by hand (quadrat) or machine are potentially higher than commercial grower-harvested yields.

For the purposes of carbon-accounting systems, HI data are required to define the amount of carbon retained in the form of crop residues at a site after harvest. It is clear that HI data used for this purpose must be reflective of typical agricultural harvesting practices and account or compensate for any leaf drop prior to harvest. Problems are likely to be more significant for noncereal crops for which leaf drop is more pronounced.

4. DRY MATTER, GRAIN YIELD, AND HARVEST INDEX VALUES FOR CROPS

Lower limits for crop dry matter, grain yield, and harvest index would all be near zero for rainfed crops, due to occasional drought or other catastrophic crop failure (e.g., pests, disease, weeds, frost). Unfortunately very low and zero crop yields are rarely reported in the scientific literature, partly because under situations of crop failure, or poor crop growth, many research plots are not harvested or are unharvestable. The high reliance on data reported in research publications within our database may skew the dataset toward greater average yields. Nevertheless, for some crops zero and very low grain yields have been reported ([Table 5](#)), and although zero yield crops would by default have zero harvest indices, this was not specifically reported for these particular crops.

Across the dataset mean harvest index for wheat and barley (0.37) was the same as that averaged for all grain legumes (0.37, lupin, field pea, faba bean, lentil, chickpea, and vetch), while oat harvest index was lowest (0.21) followed by canola (0.28). The low HI of oat may reflect its primary role as a forage crop and a low intensity of breeding for grain yield, and for canola the heavy energy (oil) investment in seeds (see [Table 1](#)). In [Fig. 7](#) we plot the cost of seed production against mean HI for crops in Australia where it can be seen that high energy cost of seed production is correlated with lower crop harvest index. Again the position of oat likely reflects a lack of breeding effort for high grain yield compared to that achieved with wheat and barley. The apparently high HI value for faba bean may partly be due to non collection of fallen leaves in HI estimations.

Table 5 Gross summary of grain crop dry matter, grain yield, and harvest index data reported for dryland crops in Australia (from [Unkovich et al., 2006](#))

Crop	Species	Shoot dry matter (kg/ha)				Grain yield (kg/ha)				Harvest index			
		Mean	<i>n</i>	Min	Max	Mean	<i>n</i>	Min	Max	Mean	<i>n</i>	Min	Max
Wheat	<i>Triticum aestivum</i>	6699	1015	140	22,250	2449	14,535	19	8600	0.37	1266	0.08	0.56
Barley	<i>Hordeum vulgare</i>	7777	96	12	19,300	2594	2992	5	8180	0.38	117	0.09	0.57
Lupin	<i>Lupinus angustifolius</i>	5148	165	714	13,200	1575	919	17	4482	0.30	229	0.04	0.5
Canola	<i>Brassica napus</i>	6829	129	51	17,793	1569	5449	20	5160	0.28	117	0.04	0.41
Oat	<i>Avena sativa</i>	9669	163	1050	28,102	2535	872	3	6840	0.21	21	0.11	0.48
Sorghum	<i>Sorghum bicolor</i>	6855	78	460	20,763	3232	109	80	10,000	0.46	78	0.14	0.7
Triticale	<i>Triticum durum</i> × <i>Secale cereale</i>	9568	6	6065	11,230	3109	409	52	7264	0.34	10	0.28	0.46
Field pea	<i>Pisum sativum</i>	4706	167	1140	11,580	1486	708	150	4535	0.36	185	0.06	0.58
Chickpea	<i>Cicer arietinum</i>	3783	180	80	10,245	1128	830	0	4033	0.37	188	0.06	0.55
Faba bean	<i>Vicia faba</i>	5130	49	1330	10,740	1727	720	163	5752	0.44	48	0.11	0.58
Lentil	<i>Lens culinaris</i>	3817	42	1440	9310	1307	261	0	4134	0.33	44	0.06	0.51
Maize	<i>Zea mays</i>	20,733	12	15,500	28,000	5659	51	1445	14,400	0.52	7	0.41	0.62
Vetch	<i>Vicia sativa</i>	3701	24	222	9890	1070	130	0	2830	0.38	23	0.16	0.47
Peanut	<i>Arachis hypogea</i>	3730	16	940	12,378	1553	50	51	6000	0.33	32	0.02	0.57

Data are from an exhaustive summary of the literature, not a rigorous strategic sampling across all regions and years. Crops are listed in decreasing order of sowing area.

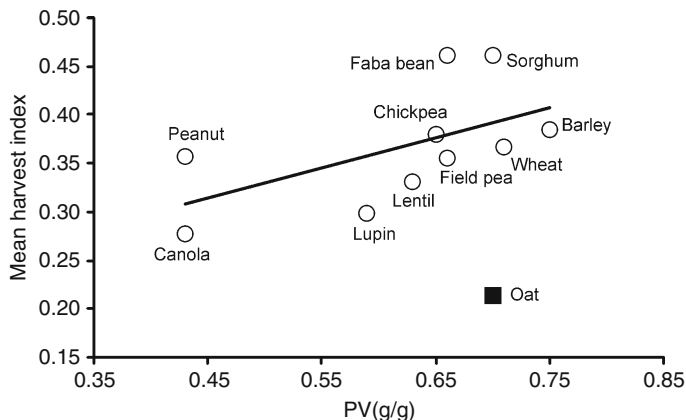


Figure 7 Scatterplot of the cost of seed production (PV, g seed/g carbohydrate) and mean harvest index for field crops in Australia. Excluding oat, the PV accounted for 33% of the variation in HI across crops ($HI = 0.3133 \times PV + 0.1725$).

In [Table 6](#) we have extracted the maximum records of dry matter, yield, and harvest index and their origins as a matter of interest. Such values are often difficult to find and may prove useful for those interested in upper physiological or agronomic limits, or modeling potential C fluxes. Reported crop dry matter accumulation maxima in Australia ([Table 6](#)) range from < 10 t/ha for a range of crop legumes (lentil, field pea, mungbean), through 10–15 t/ha for chickpea, faba bean, soybean, peanut, lupin, and triticale, 17 t/ha for canola, 21 t/ha for barley, 24 t/ha for wheat, and 28 t/ha for oat. Being irrigated, rice is the most productive of the C3 crops with a reported maxima of 31.6 t DM/ha. Among the C4 cereals reported DM maxima for grain sorghum of 22 t/ha was lower than the 28 t/ha for maize.

The upper limits of crop growth are a function of crop leaf area development, incident radiation, and radiation use efficiency ([Sinclair, 1994](#)), assuming that water, nutrients, weeds, and other pests are not limiting. Maximum economic yields are then a function of harvest index. Relatively simple crop models have been effective in accurately estimating maximum potential yields and maximum achieved yields in a given environment ([Sinclair, 1993](#)). Establishment of maxima is useful from the point of view of setting upper limits for management and for models, and perhaps for filtering out unrealistic or poor quality data.

Upper limits for HI in wheat and barley have been suggested to be *ca* 0.6, based on likely assimilate partitioning and the need for stem strength to prevent lodging, etc. ([Austin *et al.*, 1980](#); [Slafer *et al.*, 1999](#)). However, many of the HI values above 0.5 reported in the literature may be due to a reporting of apparent rather than actual harvest indices for the reasons outlined earlier. The coefficients of variation for HI ([Table 7](#)) ranged from 16% in maize to 48% in oat and 55% in peanut.

Table 6 Maximum values recorded in the literature for biological yield (DM yield, t/ha), grain yield (GY t/ha), and harvest index (HI = GY/DM yield) for a range of dryland crops in Australia

Crop	DM yield	Source	GY	Source	HI	Source
Barley	19.3 (20.9)	1 (2)	8.2	3	0.57	4
Canola	17.8	5	5.2	5	0.41	6
Chickpea	10.2 (10.5)	7 (8)	4.0	9	0.55	7
Faba_bean	10.7	10	5.8	11	0.62	12
Field_pea	11.6	9	4.5	10	0.58	11
Lentil	9.3	13	4.1	3	0.51	14
Lupin	13.2	15	4.5	16	0.50	17
Maize	28.0	18	14.4	18	0.62	18
Mungbean	6.8 (9.4)	19	2.4	19	0.55	19
Oat	28.1	20	6.8	21	0.48	22
Peanut	12.4	23	6.0	23	0.57	23
Rice	31.6	24	14.0	25	0.70	26
Sorghum	21.6	27	10.0 (12.4)	28	0.70	29
Soybean	12.7	30	4.5	30	0.35	31
Sunflower	9.0 (15.6)	31 (32)	2.9 (5.7)	30 (32)	0.48 (0.54)	31 (33)
Triticale	17.0	34	7.3	2	0.46	35
Vetch	9.9	37	2.8	37	0.47	17
Wheat	22.3 (23.9)	5 (38)	8.0 (8.9)	38	0.56	39

Data sources are indicated in the table. Note that entries for individual crops do not necessarily come from the same crops or studies. (Where maximum values for irrigated crops were greater than for dryland records, these are included in brackets for comparison). Lupin is *Lupinus angustifolius* for peanut is pod yield.

Data sources (1) Boonchoo *et al.* (1998), (2) Mitchell *et al.* (1996), (3) NVT (2008), (4) Fathi *et al.* (1997), (5) Zhang and Evans (2004), (6) Hirth *et al.* (2001), (7) Berger *et al.* (2004), (8) Leport *et al.* (1999), (9) Haigh (2006), (10) Loss and Siddique (1997), (11) Egan and Crouch (2004), (12) Loss *et al.* (1998), (13) Siddique *et al.* (1998), (14) Turner *et al.* (1996), (15) Armstrong *et al.* (1997), (16) Bishop and Mendham (1996), (17) Siddique *et al.* (2001), (18) Humphreys *et al.* (2003), (19) Thomas *et al.* (2004), (20) Wheeler and McMurray (2000), (21) Zwer and Hoppe (2004), (22) Angus and van Herwaarden (1989), (23) Bell *et al.* (1991), (24) Ockerby and Fukai (2001), (25) ten Berge *et al.* (1997), (26) Beecher *et al.* (1994), (27) Ferraris and Charles-Edwards (1986), (28) Thompson and Heenan (1989), (29) Holland and Felton (1989), (30) Institute of Sustainable Irrigated Agriculture (Vic.) (1995), (31) Herridge and Holland (1992), (32) Hall *et al.* (1989), (33) Pritchard and Small (1992), (34) Bakker (2008), (35) Zubaidi *et al.* (1999), (36) Dear *et al.* (2005), (37) McMurray and Matic (2003), (38) Stapper and Fischer (1990), (39) Angus and Fischer (1991).

Our dataset is likely to be skewed toward certain sites, years, and treatments. For example, one set of 20 entries may have come from a series of treatments or cultivars at one site in 1 year, while another single entry may represent one site, treatment, and cultivar for 1 year. Which data have the most relevance or value will depend on the specific question being asked. A sensible approach when analyzing the data might thus be to average the data on a data source $\times$ site $\times$ year basis. In this way the values would reflect variation in HI as a function of climate and soils and perhaps less due to local management. This may be particularly important where extreme

Table 7 Coefficient of variation in crop harvest index from the database across all dryland and irrigated crops, across all sites, years, and data sources

Crop	cv	<i>n</i>
Barley	24	127
Canola	23	124
Chickpea	31	205
Faba bean	25	131
Field pea	34	185
Lentil	43	44
Lupin	30	262
Maize	16	33
Mungbean	21	22
Oat	48	21
Peanut	55	67
Rice	23	141
Sorghum	22	103
Soybean	27	16
Sunflower	17	37
Triticale	18	10
Vetch	27	23
Wheat	20	1422

experimental treatments might unduly influence HI beyond what growers might achieve. The number of HI entries (*n*), the measured and fitted frequency distribution of HI and other statistics for the measured and fitted distributions are given in [Fig. 8](#) for all crops where more than 30 data points remained after averaging data on a data source $\times$ site $\times$ year basis. From this it can be seen that only seven crops have HI data from more than 30 data source $\times$ sites $\times$ year combinations, and HI data for many Australian field crops is thus somewhat scarce. Since mean values can be misleading, depending on the frequency distribution of HI, we have presented a range of statistics that describe various aspects of the distribution of HI obtained for each crop (minimum, maximum, mean, median, mode, standard deviation, skewness, and kurtosis). Each parameter describes particular aspects of the distribution of HI values. In addition to the usual statistics (minimum, maximum, mean, median, mode, and standard deviation), values for skewness and kurtosis have also been provided. Skewness values provide a measure of the lack of symmetry with negative values indicating the data are skewed left (long left tail) and positive values indicating the data are skewed right (long right tail) relative to a normal distribution with a skewness of 0. Kurtosis provides a measure of whether the data are peaked or flat relative to a normal distribution which has a Kurtosis value of 3. Datasets with a high

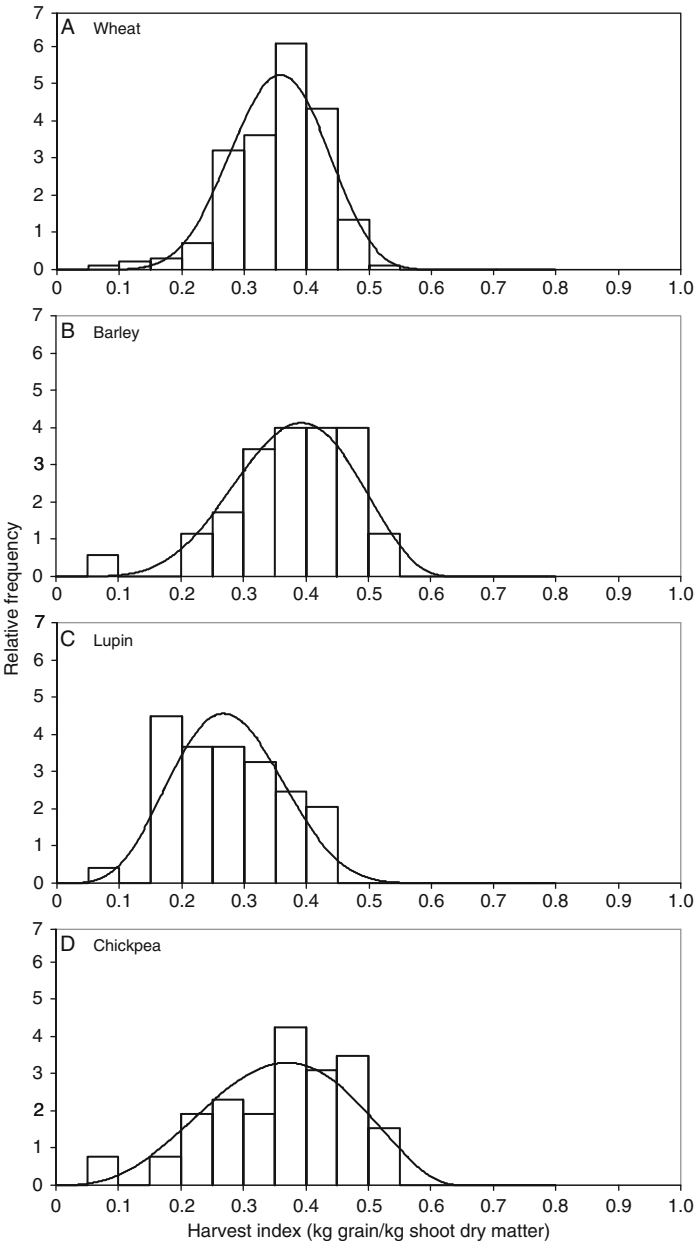


Figure 8 (Continued)

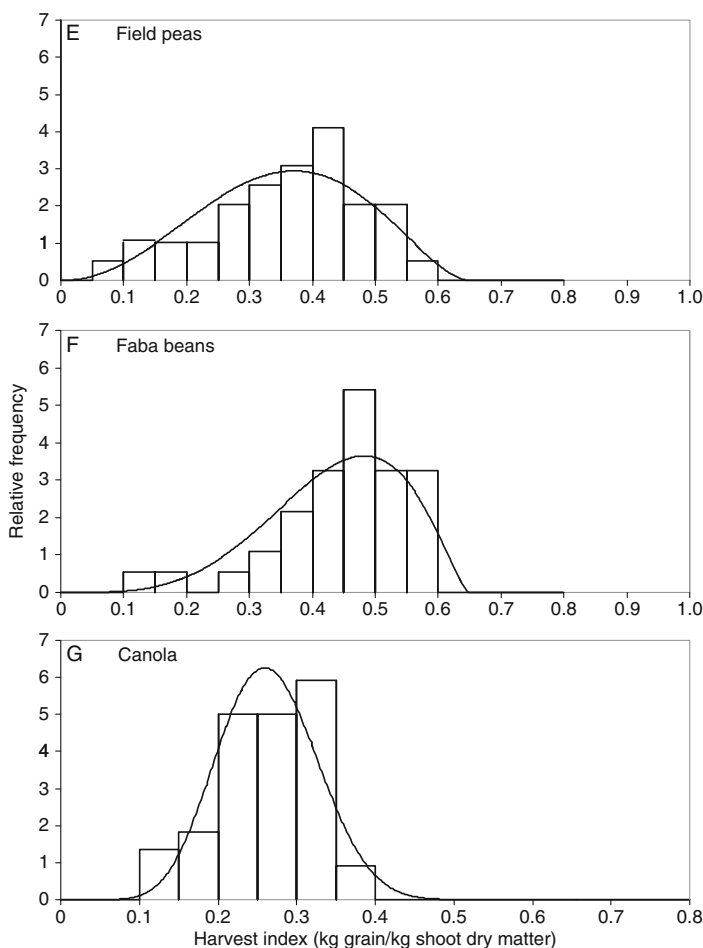


Figure 8 Measured (histogram) and fitted (line) frequency distributions for crop harvest index from the database where the number of site $\times$ year observations > 30 , that is, for wheat, barley, lupin, chickpea, field pea, faba bean, and canola. Data were first averaged by site $\times$ year and then organized into 0.05 HI interval increments. The form of the fitted distribution was BetaGeneral (α_1 , α_2 , min, and max) with the minimum and maximum values set to 0 and 0.65, respectively.

kurtosis have a distinct peak near the mean, decline rapidly, and have heavy tails. Datasets with low kurtosis have a flat top near the mean rather than a sharp peak. The values obtained for each of these statistics for the measured HI frequency distributions will be discussed in the subsequent sections where the HI data for individual crops are examined. The fitted distributions were obtained with @Risk (version 5, Palisade Corporation, Ithaca, New York)

using a BetaGeneral distribution with four fitting parameters: α_1 , α_2 , min, and max. The minimum HI (min) was set to 0 and the maximum HI (max) to 0.65, and then the values for α_1 and α_2 were fitted to the measured data. For all seven crops, the fitted frequency distributions appear to provide a reasonable estimate of the measured frequency distributions.

We follow with a brief summary of what is known of HI for each crop, in decreasing order of importance, in terms of the area sown in Australia (Unkovich *et al.*, 2009). We conclude each crop with summary of our HI data averaged on a site $\times$ year basis which we believe provides the safest interpretation given the nature and purpose of the dataset. This site $\times$ year averaged dataset can be made available to readers on request. Please contact the corresponding author.

4.1. Wheat

For wheat more than 14,500 yield estimates were obtained and >1200 for harvest index, many more than for any other crop (Table 5). Mean shoot dry matter yield for rainfed wheat in the crops reported was 6.7 t/ha with a maximum of 22.5 t/ha. Harvest index values ranged from 0.04 to 0.59. Variation in HI at a given site over time can be examined in the rotation trial at Tarlee, South Australia. Here, HI for wheat ranged from 0.23 to 0.50 (1980–1987, calculated from reports, see Schultz, 1995). Estimates of HI at Merredin (WA) sites in our database cover 11 years from various locations, ranging from 0.25 to 0.54. Over 17 years at Wagga NSW, HI of wheat has ranged from 0.08 to 0.42, and at Walpeup (Vic) HI for wheat has ranged from 0.09 to 0.50 across 13 years of data.

Maximum growth, yield, and harvest index of wheat were examined in a series of experiments on irrigated wheat in southern NSW during 1983–1985 (Stapper and Fischer, 1990a–c), and combinations of N, sowing rate/spacing, and sowing date were employed. Within the treatments and years HI varied from 0.28 to 0.51. Maximum potential grain yield in this environment with extant material was considered to be 9–10 t/ha at 12% moisture.

At Condobolin and Cowra in NSW, HI of wheat increased with later sowing, ranging from 0.28 to 0.31 for April-sown plots to 0.38–0.42 for June-sown plots (Batten *et al.*, 1999). The range for all treatments/sites was 0.21–0.46). Batten and Khan (1987a) grew wheat at Wagga over 3 years with an N and P fertilizer factorial. They related HI to N and P nutrition. The crop HI ranged from 0.31 to 0.44, but there was no attempt to relate HI to seasonal rainfall. In NSW, increasing the sowing rate and applying N fertilizer increased DM production at the expense of HI (Duggan *et al.*, 2005). In the work of Mason and Fischer (1986), at one site in southern NSW, HI in wheat varied from 0.25 to 0.42 across 3 years and six tillage treatments.

Averaging across data source $\times$ site $\times$ year provided 194 estimates of HI for rainfed wheat, a mean of 0.36, a median of 0.36, and a mode of

approximately 0.33. More than 50% of site $\times$ year HI estimates fell within the 0.3–0.4 range (Fig. 8A). The distribution of data was negatively skewed and more peaked (kurtosis >3) than a normal distribution indicating a greater frequency of values near the mean and a longer left tail (low HI values, Table 8).

4.2. Barley

Average dry matter (8.7 t/ha) and grain yield (2.6 t/ha) of barley were higher than those obtained for wheat (Table 5). Harvest index across the whole database ($n = 127$) averaged 0.38. Harvest index in barley tends to be higher than for wheat, perhaps a consequence of its lower grain N content (see Table 1), with 70% of records being between 0.35 and 0.45 (0.275–0.475, Fig. 8B). While it would thus appear to be less variable than HI in wheat (Fig. 8A), the dataset is smaller. Values range from 0.09 in 2001 at St George in Qld, to 0.57 at Northfield SA in 1990 (Fathi *et al.*, 1997). In Queensland, the HI of two barley varieties ranged from 0.31 to 0.41, depending on seasonal water supplied (Thomas and Fukai, 1995). There are few specific studies of HI in barley in Australia.

Averaging our data for barley by source $\times$ site $\times$ year gave 35 estimates of HI for rainfed barley, with a mean of 0.38, a median of 0.39, and mode of

Table 8 Statistics for measured and fitted frequency distribution for crop harvest index from the database where the number of site $\times$ year observations >30 , that is, for wheat, barley, lupin, chickpea, field pea, faba bean, and canola

	Min	Max	Mean	Mode	Median	s.d.	Skewness	Kurtosis	<i>n</i>
Wheat	0.08	0.52	0.36	0.33	0.36	0.07	−0.74	3.89	194
Fitted	0.00	0.65	0.36	0.36	0.36	0.07	−0.08	2.73	
Barley	0.10	0.53	0.38	0.47	0.39	0.09	−0.90	3.91	35
Fitted	0.00	0.65	0.38	0.39	0.39	0.09	−0.19	2.63	
Lupin	0.07	0.44	0.28	0.20	0.29	0.08	0.03	2.21	49
Fitted	0.00	0.65	0.28	0.27	0.27	0.08	0.15	2.67	
Chickpea	0.07	0.55	0.36	0.37	0.38	0.11	−0.62	2.86	52
Fitted	0.00	0.65	0.36	0.37	0.36	0.11	−0.13	2.45	
Field pea	0.06	0.56	0.36	0.42	0.37	0.12	−0.48	2.60	39
Fitted	0.00	0.65	0.36	0.37	0.36	0.12	−0.13	2.37	
Faba bean	0.11	0.58	0.45	0.52	0.46	0.11	−1.20	4.38	37
Fitted	0.00	0.65	0.44	0.48	0.45	0.10	−0.47	2.73	
Canola	0.10	0.38	0.27	0.30	0.28	0.06	−0.75	3.32	44
Fitted	0.00	0.65	0.27	0.26	0.26	0.06	0.14	2.82	

Data were first averaged by site $\times$ year and then organized into 0.05 HI interval increments. The form of the fitted distribution was BetaGeneral (α_1 , α_2 , min, and max) with the minimum and maximum values set to 0 and 0.65, respectively.

approximately 0.47 (Table 8). As noted for wheat, the distribution of HI data was negatively skewed and more peaked than a normal distribution.

4.3. Lupin

Database entries for lupin total 165 for DM yield, 919 for GY, and 229 for harvest index. Mean values across the database were 5.2 t/ha for DM yield, 1.6 t/ha for GY, and 0.30 for HI (Table 5). Maximum DM yield of 12.6 t/ha for irrigated lupin (Dracup *et al.*, 1998a) and 11.1 t/ha for rainfed lupin at Eradu, WA, have been reported. The grain yield maxima of 4.4 t/ha in the database was recorded at Cressy, TAS (Bishop and Mendham, 1996). Across the database HI of lupin ranged from 0.04 to 0.50, with an average of 0.30 (Table 5).

While the upper limit of 0.5 used in the modeling of Farre *et al.* (2004) seems sensible in the light of the aforementioned data, they included roots in their HI calculation. Large variation in lupin HI has been suggested (Dracup *et al.*, 1998a), but we can only find 49 site $\times$ year estimates of HI, not enough to provide substantial evidence to demonstrate this. Even the article of Delane *et al.* (1988), which is often cited as evidence of variable HI (e.g., Dracup and Kirby, 1996; Palta and Ludwig, 1998), concluded that “overall, results suggest that very poor pod setting on the main stem and low HI are not as widespread a problem as previously thought.” In the studies of Delane *et al.*, across 38 sites, crop dry matter explained 67–83% of yield variation of three genotypes, indicating that factors influencing pod set probably account for no more than 20% of yield variation. Using supplementary irrigation, Dracup *et al.* (1998a) showed that terminal drought can have a substantial influence on yield and HI of lupin. HI increased from 0.14 in unirrigated plants to 0.22 with the addition of ca 190 mm water. Additional water during pod fill was seen as key to increased HI in these experiments. Irrigation increased pod survival which led to the increased yield and harvest index. In lupin, mild transient water stress may also increase HI by redirecting plant resources from vegetative growth to seed fill (French and Turner, 1991). In lupin, up to 94% of flowers and pods may be aborted (Atkins and Pigeaire, 1993); however, keeping flowers on such plants does not necessarily lead to increased seed yield (Palta and Ludwig, 1996), and seed filling rather than increased pod number would appear to be more likely to increase lupin HI and yield (Palta *et al.*, 2003). Increasing C supply during pod fill (Palta and Ludwig, 2000) increased seed yield but not HI as concurrent vegetative growth maintained its share of the C sink.

Averaging across site $\times$ year provided 49 estimates of HI for lupin (Fig. 8C) with a mean of 0.28, a median of 0.29, and mode of 0.20 (Table 8). Approximately two-thirds of the HI values fell between the 0.20 and 0.30 class intervals. Little evidence for skewness existed and the distribution was flatter than a normal distribution.

4.4. Sorghum

Seventy five percent of the crop entries in the database were for rainfed production ($n = 113$). Mean DM yield for rainfed sorghum in the database was 6.8 t/ha ($n = 78$, [Table 5](#)) and 11.4 t/ha for irrigated crops ($n = 18$). Grain yield varied from 0.8 to 10.0 t/ha for rainfed crops and 2.7–12.4 t/ha for irrigated crops. Harvest index for rainfed crops ranged from 0.14 to 0.70 (mean = 0.46), and for irrigated crops 0.26–0.52 (mean = 0.43). There were insufficient data ($n = 19$) to provide meaningful insights from site $\times$ year averaging.

4.5. Chickpea

Dry matter production of chickpea in the database averages 3.8 t/ha and grain yield 1.1 t/ha. The maximum DM yield recorded was 10.7 t/ha for an irrigated crop at Merredin, WA ([Leport *et al.*, 1999](#)), with an accompanying grain yield of 4.2 t/ha. For rainfed crops the maxima were very similar at 10.2 t DM/ha (Warwick Qld [Berger *et al.*, 2004](#) and personal communication) and 4 t grain/ha at Tamworth, NSW ([Haigh, 2006](#)). More than half (141) of the 212 chickpea HI entries were from 13 varieties over five sites and 2 years provided courtesy of [Berger *et al.* \(2004\)](#) and personal communication. There were 9 years of data from Merredin (WA) with HI there varying from 0.07 to 0.54. In Queensland ([Thomas and Fukai, 1995](#)) HI of chickpea ranged from 0.19 to 0.52, depending on seasonal water supply under controlled conditions.

Across sites and years ($n = 52$) chickpea HI averaged 0.36 with a median of 0.38 and mode of 0.37 ([Table 8](#)). The distribution ([Fig. 8D](#)) was negatively skewed and approximated the normal distribution in terms of its peakedness.

4.6. Field pea

Mean dry matter production of field pea in our database is 4.7 t/ha and grain yield 1.5 t/ha. Maxima were 11.6 t/ha for dry matter (Mt Barker WA, [Armstrong and Pate, 1994a](#)) and 4.5 t/ha for grain yield (Bool Lagoon SA, [McMurray, 2005](#)). These are well below yields attained for the crop in Europe (e.g., [Jensen *et al.*, 2004](#)). For HI, absolute minima of 0.06 (Condobolin, NSW, 1986) and maxima of 0.58 (see [French and Ewing, 1989](#)) were recorded, although the latter may not have included fallen leaves. The mean across the database is 0.36 ([Table 5](#)). The data for field pea appear to be more variable than for most other crops ([Table 7](#)). [Armstrong and Pate \(1994b\)](#) showed that HI of semidwarf types were higher than taller, conventional cultivars, as has been shown for other crop species. Harvest indices of up to 0.6 were reported for their glasshouse-grown plants.

For our 39 site $\times$ year averaged data (Fig. 8E), HI ranged from 0.06 to 0.56, with 70% of values in the 0.35–0.50 class intervals. The distribution of HI values was negatively skewed (Table 8) and flatter than a normal distribution.

4.7. Faba Bean

Dry matter of faba bean in the database averages 5 t/ha and grain yield 1.7 t/ha, with absolute maxima of 10.7 t/ha for DM and 5.7 t/ha for grain (Dongara, WA see Loss and Siddique, 1997). As for field pea, HI in faba bean is quite variable (Table 7). The relatively high mean of 0.46 may be more apparent than real, with possible noncollection of fallen leaves skewing the values to the high end (see De Costa *et al.*, 1997).

For fababean grown at the Waite Institute (Adisarwanto and Knight, 1997) HI decreased with increasing plant density across a range of sowing dates. This resulted from fewer pods/plant at higher sowing densities. Harvest index ranged from 0.54 (later sowing, low density) to 0.4 (early sowing, high density). L'opez-Bellido *et al.* (2005) reviewed HI in faba bean in relation to plant density and presented values ranging from 0.25 to 0.52.

In a series of experiments with determinate and indeterminate fababean and $\pm$ irrigation in the United Kingdom (De Costa *et al.*, 1997), harvest index ranged from 0.34 to 0.68, for both irrigated, rainfed, and rain-excluded treatments. Across genotypes and water stress there was no trend in HI, excepting that where water was provided continuously through trickle irrigation, HI was lower than all other treatments. Higher availability of water increased assimilate partitioning to vegetative matter.

Averaging our database across site $\times$ year ($n = 37$) for faba bean gives a mean HI of 0.45 (Table 8), a negatively skewed distribution and a more peaked distribution (Fig. 8F).

4.8. Canola

Dry matter production of canola in our database averages 6.8 t/ha and grain yield 1.6 t/ha. The maximum rainfed grain yield of 5.16 t/ha (Zhang and Evans, 2004) was very similar to the irrigated yield of 5.2 t/ha recorded at Tamworth NSW (Wright *et al.*, 1992). Maximum dry matter production (15 t/ha) was recorded in a rainfed crop at Katanning WA (Ward *et al.*, 2002). Across our database HI for canola averaged 0.28 ($n = 117$), ranging from 0.04 (Thomas, 2005) to 0.41 (Hirth *et al.*, 2001).

In field experiments examining canola response to N (Hocking *et al.*, 1997), HI varied from 0.27 to 0.30 at one site in NSW, and from 0.29 to 0.34 at another. Nitrogen nutrition would thus not appear to exert a large influence on Canola HI. Robertson *et al.* (2004) examined HI in canola in response to sowing date. They found that, aside from very early or very late

(after mid-June) sowings, HI was relatively stable and yield principally a function of dry matter accumulation, with $\text{yield} = 0.335 \times \text{DM} - 27.5$ ($R^2 = 0.93$), very similar to earlier work in WA (slope 0.35 and $\times$ intercept 96 g/m²). In our database 66% of values fall into the 0.25 and 0.30 class intervals (Fig. 8G), indicating that HI in canola is fairly constrained.

Only 44 site $\times$ year records for HI of canola were found. The low mean value of 0.27 is to a large extent a consequence of the large investment in oil and protein in seed ($\text{PV} = 0.43$ g/g, see Table 1). Median and mode HI values of 0.28 and 0.30 (Table 8) were obtained and the frequency distribution (Fig. 8G) was negatively skewed and more peaked than a normal distribution. Variation in HI of canola was less than that of most other field crops examined (Table 7).

4.9. Rice

Rice crop grain yields in Australia are among the highest in the world, with the maximum reported here (14 t/ha) being close to the potential for this crop (*ca* 15 t/ha Sheehy *et al.*, 2004). Crop dry matter yield peaked at 31.6 t/ha at Walkamin, Qld (Ockerby and Fukai, 2001), or in a more traditional rice-growing region 27.5 t/ha (Yanco, NSW Horie *et al.*, 1997). Across our database HI ranges from 0.01 (Williams and Angus, 1994) to 0.71 (Dunn and Beecher, 1994). The low protein and lipid content of rice may serve to increase its harvest index. All rice grown in Australia is irrigated, and it has been shown that water deficits can be managed to increase harvest index, especially for high N crops (Zhang and Yang, 2004). If senescence of the rice crop is delayed a long, slow grain-filling period ensues, reducing HI, this is typical under high N fertility but can be circumvented by a controlled soil drying. Low temperatures during reproductive development can also have a substantial impact on HI, but this is controlled to some extent by judicious floodwater management which can effect increases in canopy temperatures (Williams and Angus, 1994).

Treatments within years can have a substantial effect on HI in rice, especially N application, for example, HI for rice varied from 0.01 to 0.63 in a single experiment in 1 year (Williams and Angus, 1994) as a function of sowing date (26 Sept. or 31 Oct.), N rate (0–250 kg N), and floodwater depth (5 or 20 cm). There was a strong negative interaction between floodwater depth and N rate. Of course such a range in environmental conditions would not normally be experienced for commercial crops which would be expected to be sown near a single optimum time and with optimized fertilizer rates. The experimental data showed that a combination of temperature and plant N content (kg/ha) at microspore development could be used to model harvest index quite effectively with an R^2 of 0.54–0.77, depending on cultivar. HI values >0.6 have been reported in the literature, for example, 0.47–0.71 (Dunn and Beecher, 1994) and

0.5–0.6 (Williams and Angus, 1994), but these were more likely to be extreme examples of what can be induced experimentally rather than what is likely in commercial crops. The maximum HI of 0.40–0.53 (depending on cultivar) in the model of Williams and Angus (1994) seems realistic.

There were only seven site $\times$ year combinations for rice in our database, insufficient to present frequency distribution data. Mean site $\times$ year HI values ranged from 0.28 at Deniliquin NSW to 0.64 in Yanco NSW, averaging 0.40.

4.10. Sunflower

The database includes only 28 records for sunflower for which the mean DM yield is 7.8 t/ha, GY 1.4 t/ha, and HI 0.4. Variation in HI is only minor with all values falling between 0.32 and 0.54 ($n = 25$). Records of maximum DM yield and GY for rainfed crops were 15.6 and 5.7 t/ha, respectively. For an irrigated crop at Gatton Qld, sown at optimum time and provided with luxuriant nutrients, total dry matter production was 17.1 t/ha at anthesis and 24.2 t/ha at maturity. Grain yield (0% moisture) was 6.02 t/ha and harvest index 0.4. Average yields for irrigated crops in the region were 1.8–3.0 t/ha (Bange *et al.*, 1977). Connor *et al.* (1985) studied variation in harvest index as a function of the timing and quantity of irrigation. Harvest index increased where water was predominantly applied postanthesis, with postanthesis irrigated crops having an HI of 0.53, cf the unirrigated crop HI of 0.38. Across the 2 years of experimentation and 12 irrigation treatments, HI for sunflower varied between 0.32 and 0.53, averaging 0.44. In India Nanja Reddy *et al.* (2003) reported HI in sunflower to vary between 0.14 and 0.44 across genotypes.

4.11. Lentil

The lentil database consists of 42 entries for DM yield, 261 grain yield estimates and 44 harvest index entries. Dry matter yield averaged 3.8 t/ha and grain yield 1.3 t/ha. Maximum DM yield of 9.3 t/ha was reported at Cunderdin, WA (Siddique *et al.*, 1998), and grain yield 4.1 t/ha at Laura, SA. (NVT, 2008). Studying a wide range of germplasms from across the globe, from wild types, through old and modern cultivars Whitehead *et al.* (2000) found HI in Lentil to reach 0.52, including fallen leaves, etc. Maximum dry matter production was 18.3 t/ha and crop residues of 14.1 t/ha were recorded across all the cultivars studied, and seed yield > 4.2 t/ha. In a glasshouse study (Shrestha *et al.*, 2006) harvest index ranged from 0.36 to 0.45 for water-stressed lentil, to 0.53–0.59 for well-watered lentil.

Our data across site $\times$ year provided only 11 HI estimates for lentil, ranging from 0.06 to 0.50, insufficient to present frequency distribution

data. The mean, median, and mode of these data were 0.33, 0.35, and 0.35, respectively.

4.12. Maize

The database includes only 33 records for harvest index in maize, with 40 DM yield estimates and 117 grain yield estimates. The mean values across the database were 19.3 t/ha (DM yield), 5.5 t/ha (GY), and 0.49 (HI), including both rainfed and irrigated crops. The mean HI was very close to that observed in the United States (Prince *et al.*, 2001). Maximum maize DM yield in the database was 28.0 t/ha and GY 20.5 t/ha (Birch *et al.*, 2006). Maize grown in temperate environments tends to have a higher HI than maize grown in tropical environments (Fischer and Palmer, 1983) due to a shorter grain-filling period in the latter, caused by water limitation.

4.13. Peanut

Across our database, there were 113 records of pod yield for peanut (60 rainfed and 53 irrigated), 56 DM yield records (34 rainfed), and 62 HI estimates (40 rainfed). Shoot dry matter for peanut in the database ranged from 0.9 t/ha for a rainfed crop to 13.3 t/ha for an irrigated crop (Bell *et al.*, 1991). Pod yields ranged from 0.51 (rainfed) to 6 t/ha for irrigated peanut. Mean values across the database for irrigated peanut were DM yield, 7.0 t/ha, pod yield 2.2 t/ha and HI 0.30. For rainfed peanut, the values were 5.5 t/ha (DM yield), 2.0 t/ha (GY), and 0.36 for HI. In one study under irrigated conditions DM yield ranged from 9 to 14.2 t/ha and HI from 0.33 to 0.56, depending on cultivar (Bell *et al.*, 1991). Averaging our data across site $\times$ year produced only five HI data points, insufficient to warrant more detailed presentation.

4.14. Oilseed poppies

Little data were available for oilseed poppies (*Papaver somniferum*), an important crop in Tasmania (10–15,000 ha) grown for the production of the opiate alkaloids, morphine, codeine, and thebaine, and which accounts for > 30% of all broadacre crop sowings in Tasmania (Unkovich *et al.*, 2009). Detailed data are kept as commercial “in confidence,” and the ABS AgStats (Australian Bureau of Statistics, 2002) does not contain yield information.

Straw harvest data available from the International Narcotics Control Board (INCB: <http://www.incb.org/>) include only the seed capsule (not seed) and a small amount of attached stalk. This only represents about 50% of the dry matter harvested, or about 25% of total crop dry matter (P. Jolly, Tasmanian Alkaloids Pty Ltd, personal communication). Some of the harvested seed may enter the culinary market (morphine), or be used as furnace fuel (thebaine) in opiate production. Poppy seed makes up about 25% of

total crop dry matter. After opiate extraction the waste capsule/straw biomass may be returned to the field for disposal. We have been unable to find direct measurement data to substantiate these approximations.

➤ **5. CROP HARVEST INDICES AND C ACCOUNTING**

Fixed harvest indices have been used to estimate net primary productivity or C inputs of croplands from statistical data on crop yields in the United Kingdom (Adger and Subak, 1996), across Europe (Vleeshouwers and Verhagen, 2002), the United States (Johnson *et al.*, 2006; Prince *et al.*, 2001), and Canada (Bolinder *et al.*, 2007). Some of these harvest indices used are given in Table 9, along with the average (site × year) values from the present Australian database for rainfed crops for comparison. Interestingly there were very significant differences in the HI values for some crops in the two United States studies highlighted in Table 9, for example, sunflower values were 0.27 (Prince *et al.*, 2001) and 0.40 (Johnson *et al.*, 2006) and oat was given as 0.52 (Prince *et al.*, 2001) or 0.44 (Johnson *et al.*, 2006). Smil (1999) estimated global crop residue resources using single HI values of 0.40 for cereals and 0.49 for legumes. With the exception of sunflower and grain sorghum, the values in our current dataset tend to be lower than those used by Prince *et al.* (2001), Johnson *et al.* (2006), and Bolinder *et al.* (2007) (Table 9). Our HI review indicates that under Australian conditions a higher quantity of residues would be returned per unit production than noted in other regions of the world. In addition, none of the abovementioned studies considered variation in HI or included such an exhaustive assessment of crop HI. The approach of using a single value for HI belies the substantial variation that exists in HI of

Table 9 Crop harvest indices used in the studies of Prince *et al.* (2001) and Johnson *et al.* (2006) in the United States and Bolinder *et al.* (2007) in Canada, along with mean values from the Australian database (averaged site × year)

Crop	From United States (Prince <i>et al.</i> , 2001)	From United States (Johnson <i>et al.</i> , 2006)	From Canada (Bolinder <i>et al.</i> , 2007)	Australian means
Corn grain	0.53	0.53	0.50	0.52
Soybean ^a	0.42	0.46	0.40	0.24
Oats	0.52	0.44	0.53	0.30
Barley	0.50	0.50	0.53	0.38
Wheat	0.50	0.45	0.40	0.36
Sunflower	0.27	0.40	—	0.40
Sorghum	0.44	0.47	0.25	0.45

^a Australian data are for *n* = 3 only, from Herridge and Holland (1992).

field crops in Australia which would create significant uncertainty in C balances. Developing an understanding of the factors operating to cause variation in HI of field-grown crops may enable HI to be varied according to local and seasonal factors and increase the accuracy of estimated crop C balances. For some crops, such as canola, which appear to have less variable HI, using a single value for C-accounting purposes may be adequate, but for many other crops, including wheat, the affect of seasonal and regional variation in HI on C accounting needs to be assessed as a potential source of error in Australia. As harvest index is known to vary as a function of environment (Prihar and Stewart, 1990) and some crops are grown across a large range of environments in Australia, it is not surprising that such variation in HI is observed.

6. SUMMARY AND CONCLUSIONS

The efficiency with which crops produce grain from dry matter is the first determinant of harvest index. In this context, cereals, which have the highest carbohydrate content, tend to have the highest harvest indices. The high energy cost of lipid production makes species with a high oil content (canola, soybean, peanut) inefficient at producing grain from assimilate, with high protein seeded species (pea, lentil, chickpea) being intermediate. For most grain crops in Australia, a harvest index rarely exceeds 0.5, excepting where fallen leaves are not taken into account in which case apparent HI may exceed 0.6. Certainly values above 0.5 should be carefully scrutinized in Australia, especially for grain legumes and oilseeds. In more favorable environments where the grain-filling period is cooler (e.g., some parts of Europe and New Zealand), harvest indices above 0.5 may be more readily achieved than in Australia.

The crop yields and dry matter data summarized here may be skewed toward higher values compared with commercial crops due to probable higher average inputs on experimental sites than commercial crops, higher grain losses during harvest of commercial crops, and the range of sites for which detailed data are available not being equally distributed across the crops sowing regions. However, with HI being a ratio and not an absolute amount our dataset may more closely represent HI of crops under commercial growing conditions.

Delayed sowing reduces the vegetative phase of crop growth and increases the harvest index for most crops, as the reproductive phase then becomes proportionately longer. Predicting or modeling crop yield or harvest index responses to temperature extremes during the reproductive phase is problematic due to differences among genotypes, and an inability to

accurately model both actual ambient temperatures and to reliably quantify grain damage from temperature extremes.

Clear relationships between fractional water use after anthesis and crop harvest index have been observed for glasshouse plants; however, while a theoretical framework has been provided for fractional postanthesis water use impacting on crop harvest index, unequivocal field data illustrating a pivotal role for profligate early water use reducing crop harvest index are lacking. Relations between HI and the fraction of water transpired after anthesis observed in controlled experiments may thus not be easily seen in field crops since other mitigating factors are not controlled, and furthermore, a complex interactions of cultivar, site, season, and management may dampen variation in crops HI. It is clear that crop nitrogen supply is implicated in variable harvest index of wheat and barley. Gross changes in HI might be able to be predicted for determinate crops from an assessment of matching climatic and soil fertility data. For indeterminate crops this challenge will be greater due to concurrent vegetative and reproductive growth and development.

Crop harvest index is a critical term in C-accounting systems for agriculture and we present here the first thorough review of variation in crop harvest indices. It must be stressed that the data presented in the present analysis were a sample from the research literature, not from all crops in all seasons. As such they represent a biased sample based on where and when research plots are located and not a random sample of all Australian crops, or even a strategic sampling or survey of Australian commercial crops. We will continue to build our database and are conducting an analysis of the variation in crop HI in our dataset as a function of seasonal climatic conditions, with a view to finding relationships between local climate and HI which may be used to vary HI temporally and spatially and improve C-accounting practices for Australian agriculture.

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REFERENCES

- Adcock, D. (2006). Soil water and nitrogen dynamics of farming systems on the upper Eyre Peninsula, South Australia. PhD, University of Adelaide, Adelaide.
- Adcock, D., McNeill, A. M., McDonald, G. K., and Armstrong, R. D. (2007). Subsoil constraints to crop production on neutral and alkaline soils in south-eastern Australia: A review of current knowledge and management strategies. *Aust. J. Exp. Agric.* **47**, 1245–1261.

- Adger, W. N., and Subak, S. (1996). Estimating above-ground carbon fluxes from UK agricultural land. *Geogr. J.* **162**, 191–204.
- Adisarwanto, T., and Knight, R. (1997). Effect of sowing date and plant density on yield and yield components in the faba bean. *Aust. J. Agric. Res.* **48**, 1161–1168.
- Allen, H. M., Pumpa, J. K., and Batten, G. D. (2001). Effect of frost on the quality of samples of Janz wheat. *Aust. J. Exp. Agric.* **41**, 641–647.
- Angus, J., and Fischer, R. A. (1991). Grain and protein responses to nitrogen applied to wheat growing on a red earth. *Aust. J. Agric. Res.* **42**, 735–746.
- Angus, J., and van Herwaarden, A. (1989). Comparative productivity of oilseeds and cereals. In “5th Australian Agronomy Conference” (M. W. Perry, Ed.), p. 525. Aust. Soc. Agron., Perth, Western Australia.
- Angus, J., and van Herwaarden, A. (2001). Increasing water use and water use efficiency in dryland wheat. *Agron. J.* **93**, 290–298.
- Armstrong, E. L., and Pate, J. S. (1994a). The field pea crop in S.W. Australia I. Patterns of biomass production and photosynthetic performance in genotypes of contrasting morphology. *Aust. J. Agric. Res.* **45**, 1347–1362.
- Armstrong, E. L., and Pate, J. S. (1994b). The field pea crop in S.W. Australia II. Effects of contrasting morphology and environment on reproductive performance. *Aust. J. Agric. Res.* **45**, 1363–1378.
- Armstrong, E. A., Heenan, D. P., Pate, J. S., and Unkovich, M. J. (1997). Nitrogen benefits of lupins, field pea and chickpea to wheat production in south-eastern Australia. *Aust. J. Agric. Res.* **48**, 39–47.
- Atkins, C., and Pigeaire, A. (1993). Application of cytokinins to flowers to increase pod set in *Lupinus angustifolius* L. *Aust. J. Agric. Res.* **44**, 1799–1819.
- Austin, R., Bingham, J., Blackwell, R., Evans, L. T., Ford, M., Morgan, C., and Taylor, M. (1980). Genetic improvement in winter wheat yields since 1900 and associated physiological changes. *J. Agric. Sci. (Camb.)* **94**, 675–689.
- Australian Bureau of Statistics. (2002). AgStats on the GSP Australian Bureau of Statistics, Canberra. Database on compact disc.
- Bakker, D. (2008). Crop establishment and productivity with improved root zone drainage. In “GRDC Farming Systems Crop Updates” (D. Abrecht, Ed.), pp. 89–93. Agriculture WA, Perth.
- Bange, M., Hammer, G., and Rickert, K. (1977). Environmental control of potential yield of sunflower in the subtropics. *Aust. J. Agric. Res.* **48**, 231–240.
- Batten, G., and Khan, M. (1987a). Uptake and utilisation of phosphorus and nitrogen by bread wheats grown under natural rainfall. *Aust. J. Exp. Agric.* **27**, 405–410.
- Batten, G., and Khan, M. A. (1987b). Effect of time of sowing on grain yield, and nutrient uptake of wheats with contrasting phenology. *Aust. J. Exp. Agric.* **27**, 881–887.
- Batten, G., Fettell, N., Mead, J., and Khan, M. (1999). Effect of sowing date on the uptake and utilisation of phosphorus by wheat (cv. Osprey) grown in central New South Wales. *Aust. J. Exp. Agric.* **39**, 161–170.
- Beecher, H. G., Thompson, J. A., Bacon, P. E., and Heenan, D. P. (1994). Effect of cropping sequences on soil nitrogen levels, rice growth, and grain yields. *Aust. J. Exp. Agric.* **34**, 977–986.
- Bell, M., Harch, G., and Wright, G. (1991). Plant population studies on peanut (*Arachis hypogaea* L.) in subtropical Australia 1. Growth under fully irrigated conditions. *Aust. J. Exp. Agric.* **31**, 535–543.
- Berger, J., Turner, N., Siddique, K. H. M., Knights, E., Brinsmead, R., Mock, I., Edmonson, C., and Khan, T. (2004). Genotype by environment studies across Australia reveal the importance of phenology for chickpea (*Cicer arietinum* L.) improvement. *Aust. J. Agric. Res.* **55**, 1071–1084.
- Birch, C., McLean, G., and Sawers, A. (2006). Analysis of high yielding maize production—A study based on a commercial crop. In ““Water to Gold” 6th Triennial Conference”

- (E. Humphreys, K. O'Keefe, N. Hutchins, and R. Gill, Eds.), pp. 55–65. Maize Association of Australia, Griffith.
- Bishop, A. C., and Mendham, N. J. (1996). Quantifying the yield-density relationship for narrow-leaved lupin (*Lupinus angustifolius*) in Tasmania. In "8th Australian Agronomy Conference" (M. Asghar, Ed.), pp. 100–103. Australian Society of Agronomy, Toowoomba.
- Boer, R., Campbell, L. C., and Fletcher, D. J. (1993). Characteristics of frost in a major wheat-growing region of Australia. *Aust. J. Agric. Res.* **44**, 1731–1743.
- Bolinder, M. A., Angers, D. A., and Dubuc, J. P. (1997). Estimating shoot to root ratios and annual carbon inputs in soils for cereal crops. *Agric. Ecosyst. Environ.* **61–66**, 61.
- Bolinder, M. A., Janzen, H. H., Gregorich, E. G., Angers, D. A., and VandenBygaart, A. J. (2007). An approach for estimating net primary productivity and annual carbon inputs to soil for common agricultural crops in Canada. *Agric. Ecosyst. Environ.* **118**, 29–42.
- Bondeau, A., Smith, P. C., Zaehle, S., Schaphoff, S., Lucht, W., Cramer, W., Gerten, D., Lotze-Campen, H., Muller, C., Reichstein, M., and Smith, B. (2007). Modelling the role of agriculture for the 20th century global terrestrial carbon balance. *Glob. Change Biol.* **13**, 679–706.
- Boonchoo, S., Fukai, S., and Hetherington, S. E. (1998). Barley yield and grain protein concentration as affected by assimilate and nitrogen availability. *Aust. J. Agric. Res.* **49**, 695–706.
- Calderini, D. F., Reynolds, M. P., and Slafer, G. A. (1999). Genetic gains in wheat yield and associated physiological changes during the twentieth century. In "Wheat ecophysiology and yield determination" (E. Satorre and G. Slafer, Eds.), pp. 351–377. Food Products Press, New York.
- Cantero-Martínez, C., O'Leary, G., and Connor, D. (1995). Stubble retention and nitrogen fertilisation in a fallow-wheat rainfed cropping system. 1. Soil water and nitrogen conservation, crop growth and yield. *Soil Till. Res.* **34**, 79–94.
- Cassels, J., and Armstrong, E. L. (1998). Timing all important in boosting quality. *Farming Ahead* **81**, 63–64.
- Connor, D., Jones, T., and Palta, J. (1985). Response of sunflower to strategies of irrigation I. Growth, yield and the efficiency of water use. *Field Crops Res.* **10**, 15–36.
- Constable, G. A., and Gleeson, A. C. (1977). Growth and distribution of dry matter in cotton (*Gossypium hirsutum* L.). *Aust. J. Agric. Res.* **28**, 249–256.
- Cornish, P., and Murray, G. (1989). Low rainfall rarely limits wheat yields in southern New South Wales. *Aust. J. Exp. Agric.* **29**, 77–83.
- Croser, J. S., Clarke, H. J., Siddique, K. H. M., and Khan, T. N. (2003). Low-temperature stress: Implications for Chickpea (*Cicer arietinum* L.) improvement. *Crit. Rev. Plant Sci.* **22**, 185.
- Dang, Y. P., Dalal, R. C., Routley, R., Schwenke, G. D., and Daniells, I. (2006). Subsoil constraints to grain production in the cropping soils of the north-eastern region of Australia: An overview. *Aust. J. Exp. Agric.* **46**, 19–35.
- De Costa, W. A. J. M., Dennett, M. D., Ratnaweera, U., and Nyalemegbe, K. (1997). Effects of different water regimes on field-grown determinate and indeterminate faba bean (*Vicia faba* L.). II. Yield, yield components and harvest index. *Field Crops Res.* **52**, 169–178.
- Dear, B. S., Kaiser, A., and Piltz, J. (2005). Yield and digestibility of legume and oat forages, Department of Primary Industries (NSW), Primefact 52.
- Delane, R. J., Matthews, J. C., and Hamblin, J. (1988). Effect of genotype and environment on harvest index of *L. angustifolius*. In "Proceedings 5th International Lupin Conference, Poznan, Poland", p. 527.
- Donald, C. M., and Hamblin, J. (1976). The biological yield and harvest index of cereals as agronomic and plant breeding criteria. *Adv. Agron.* **28**, 361–405.

- Dracup, M. (1994). Physiological aspects to lupin improvement for water-limited Mediterranean environments. In "First Australian Lupin Technical Symposium" (M. Dracup and J. Palta, Eds.), pp. 140–151. Agriculture Western Australia, Perth.
- Dracup, M., and Kirby, E. J. M. (1996). Pod and seed growth and development of narrow-leaved lupin in a water limited Mediterranean-type environment. *Field Crops Res.* **48**, 209–222.
- Dracup, M., Reader, M., and Palta, J. (1998a). Variation in yield of narrow-leaved lupin caused by terminal drought. *Aust. J. Agric. Res.* **49**, 799–810.
- Dracup, M., Turner, N. C., Tang, C., Reader, M., and Palta, J. (1998b). Responses to abiotic stresses. In "Lupins as Crop Plants: Biology, Production and Utilisation" (J. S. Gladstones, C. A. Atkins, and J. Hamblin, Eds.), CABI International, Wallingford.
- Duggan, B., Richards, R., van Hewarden, A., and Fettel, N. (2005). Agronomic evaluation of a tiller inhibition gene (tin) in wheat. I. Effect on yield, yield components, and grain protein. *Aust. J. Agric. Res.* **56**, 169–178.
- Dunn, B. W., and Beecher, H. G. (1994). Green manuring legume pasture for aerial-sown rice. *Aust. J. Exp. Agric.* **34**, 967–975.
- Duthion, C., and Pigeaire, A. (1993). Stability of the nitrogen concentration of inoculated white lupins during pod development. *Ann. Bot.* **72**, 55–61.
- Egan, J., and Crouch, J. (2004). Faba bean—2004 variety trials. In "South Australian Field Crop Evaluation Program 2004/2005" (R. Wheeler and L. McMurray, Eds.), p. 9. Primary Industries and Resources, South Australia, Adelaide.
- Egan, J., McCormack, P., Bennie, M., and McEvoy, K. (1994). Grain legumes for low rainfall areas. "Harvest and Livestock Report for Upper Eyre Peninsula 1993/94", pp. 35–37. Primary Industries South Australia, Port Lincoln.
- Erskine, W., Nassib, A., and Telaye, A. (1988). Breeding for morphological traits. In "World crops: Cool season food legumes" (R. J. Summerfield, Ed.), pp. 117–127. Kluwer Academic, Dordrecht.
- Evans, L. T. (1993). Crop evolution, adaptation and yield. Cambridge University Press, Cambridge.
- Farre, I., Robertson, M., Asseng, S., French, R., and Dracup, M. (2004). Simulating lupin development, growth, and yield in a Mediterranean environment. *Aust. J. Agric. Res.* **55**, 863–877.
- Fathi, G., McDonald, G., and Lance, R. (1997). Effects of post-anthesis water stress on the yield and grain protein concentration of barley grown at two levels of nitrogen. *Aust. J. Agric. Res.* **48**, 67–80.
- Ferraris, R., and Charles-Edwards, D. (1986). A comparative analysis of the growth of sweet and forage sorghum crops. I Dry matter production, phenology and morphology. *Aust. J. Agric. Res.* **37**, 495–512.
- Fischer, R. A. (1979). Growth and water limitation to dryland wheat yield in Australia: A physiological framework. *J. Aust. Inst. Agric. Sci.* **45**, 83–94.
- Fischer, K. S., and Palmer, A. (1983). Maize. In "Potential productivity of field crops under different environments" (Anon, Ed.), pp. 155–180. International Rice Research Institute, Los Banos, Philippines.
- Flood, R., and Martin, P. (2001). Nitrogen accumulation and distribution at anthesis and maturity in ten wheats grown at three sites in north-western Victoria. *Aust. J. Exp. Agric.* **41**, 533–540.
- French, R. J., and Ewing, M. A. (1989). Soil type influences the relative yields of different cereals and crop legumes in the Western Australian wheatbelt. *Aust. J. Exp. Agric.* **29**, 829–835.
- French, R. J., and Turner, N. C. (1991). Water deficits change dry matter partitioning and seed yield in narrow-leaved lupins (*Lupinus angustifolius* L.). *Aust. J. Agric. Res.* **42**, 471–484.

- Gomez-Macpherson, H., and Richards, R. A. (1995). Effect of sowing time on yield and agronomic characteristics of wheat in south-eastern Australia. *Aust. J. Agric. Res.* **46**, 1381–1399.
- Goyne, P., Milroy, S., Lilley, J., and Hare, J. (1993). Radiation interception, radiation use efficiency and growth of barley cultivars. *Aust. J. Agric. Res.* **44**, 1351–1366.
- Greenwood, E. A., Farrington, P., and Beresford, J. D. (1975). Characteristics of the canopy, root system and grain yield of a crop of *Lupin angustifolius* cv. Unicrop. *Aust. J. Agric. Res.* **26**, 497–510.
- Gregory, P. J., Palta, J. A., and Batts, G. R. (1995). Root systems and root:mass ratio–carbon allocation under current and projected atmospheric conditions in arable crops. *Plant Soil* **187**, 221–228.
- Haigh, B. (2006). Personal communication. NSW Agriculture, Tamworth.
- Hall, A. J., Connor, D. J., and Whitfield, D. M. (1989). Contribution of pre-anthesis assimilates to grain-filling in irrigated and water-stressed sunflower crops I. Estimates using labelled carbon. *Field Crops Res.* **20**, 95–112.
- Hamblin, J., Delane, R., Bishop, A., and Gadstones, J. Y. (1986). Yield potential of reduced branching lupins (*Lupinus angustifolius*) on sandy soils in a short season environment. *Aust. J. Agric. Res.* **37**, 611–620.
- Hardie, D. C., Baker, G. J., and Marshall, D. R. (1995). Field screening of *Pisum* accessions to evaluate their susceptibility to the pea weevil (Coleoptera: Bruchidae). *Euphytica* **84**, 155–161.
- Herridge, D. F., and Holland, J. F. (1992). Production of summer crops in northern New South Wales. I. Effects of tillage and double cropping on growth, grain and N yields of six crops. *Aust. J. Agric. Res.* **43**, 105–122.
- Hirth, J., Haines, P., Ridley, A. M., and Wilson, K. (2001). Lucerne in crop rotations on the Riverine Plains. 2. Biomass and grain yields, water use efficiency, soil nitrogen, and profitability. *Aust. J. Agric. Res.* **52**, 279–293.
- Hocking, P., Randall, P., and DeMarco, D. (1997). The response of dryland canola to nitrogen fertilizer: Partitioning and mobilization of dry matter and nitrogen, and nitrogen effects on yield components. *Field Crops Res.* **54**, 201–220.
- Hodgson, A. S. (1978). Rapeseed adaptation in northern New South Wales. II Predicting plant development of *Brassica campestris* L. and *Brassica napus* L. and its implications for planting time, designed to avoid water deficit and frost. *Aust. J. Agric. Res.* **29**, 711–726.
- Hoffman, T. K., and Kolb, F. L. (1998). Effects of barley yellow dwarf virus on yield and yield components of drilled winter wheat. *Plant Dis.* **82**, 620–624.
- Hooda, R. S., Rao, A. S., Luthra, Y. P., Sheoran, I. S., and Singh, R. (1986). Partitioning and utilization of carbon and nitrogen for dry matter and protein production in chickpea (*Cicer arietinum* L.). *J. Exp. Bot.* **37**, 1492–1502.
- Holland, J. F., and Felton, W. L. (1989). Effect of tillage practice on grain sorghum production in northern New South Wales. *Aust. J. Exp. Agric.* **29**, 843–848.
- Horie, T., Ohnishi, M., Angus, J. F., Lewin, L. G., Tsukaguchi, T., and Matano, T. (1997). Physiological characteristics of high-yielding rice inferred from cross-location experiments. *Field Crops Res.* **52**, 55–67.
- Huhn, M. (1990a). Comments on the calculation of mean harvest indices. *J. Agron. Crop Sci.* **165**, 86–93.
- Huhn, M. (1990b). On the variability of harvest indices. *J. Agron. Crop Sci.* **164**, 271–281.
- Humphreys, E., Smith, D. J., White, R. J., Xevi, E., and Godwin, D. C. (2003). Maizeman—A decision support system for water and nitrogen management. In “5th Australian Maize Conference” (C. J. Birch and S. R. Wilson, Eds.), pp. 131–134. Maize Association of Australia, Toowoomba.
- Institute of Sustainable Irrigated Agriculture (Vic). (1995). Biennial Research Report (1994–1995). Tatura.

- Jackson, R. B., Canadell, J. J., Mooney, H. O., and Schulze, E. (1996). A global analysis of root distributions for terrestrial biomes. *Oecologia* **108**, 389–411.
- Jensen, C. R., Joernsgaard, B., Andersen, M. N., Christiansen, J. L., Mogensen, V. O., Friis, P., and Petersen, C. T. (2004). The effect of lupins as compared with peas and oats on the yield of the subsequent winter barley crop. *Eur. J. Agron.* **20**, 405–418.
- Johnson, J. M. F., Allmaras, R. R., and Reicosky, D. C. (2006). Estimating source carbon from crop residues, roots and rhizodeposits using the National grain-yield database. *Agron. J.* **98**, 622–636.
- Kirkegaard, J., Lilley, J., Howe, G., and Graham, N. (2007). Impact of subsoil water use on wheat yield. *Aust. J. Agric. Res.* **58**, 303–315.
- Lenzen, A. W., Johnson, G. D., and Carlson, G. R. (2007). Cropping sequence and tillage system influences annual crop production and water use in semiarid Montana, USA. *Field Crops Res.* **100**, 32–43.
- Leport, L., Turner, N. C., French, R. J., Barr, M. D., Duda, R., Davies, S. L., Tennant, D., and Siddique, K. H. M. (1999). Physiological responses of chickpea genotypes to terminal drought in a Mediterranean-type environment. *Eur. J. Agron.* **11**, 279–291.
- Lobell, D. B., Hicke, J. A., Asner, G. P., Field, C. B., Tucker, C. J., and Los, S. O. (2002). Satellite estimates of productivity and light use efficiency in United States agriculture, 1982–98. *Glob. Change Biol.* **8**, 722–735.
- L'opez-Bellido, F., L'opez-Bellido, L., and L'opez-Bellido, R. J. (2005). Competition, growth and yield of faba bean (*Vicia faba* L.). *Eur. J. Agron.* **23**, 359–378.
- Lopez Pereira, M., Trapani, N., and Sadras, V. O. (2000). Genetic improvement of sunflower in Argentina between 1930 and 1995: Part III. Dry matter partitioning and grain composition. *Field Crops Res.* **67**, 215–221.
- Loss, S. P., and Siddique, K. (1997). Adaptation of faba bean (*Vicia faba* L.) to dryland Mediterranean-type environments I. Seed yield and yield components. *Field Crops Res.* **52**, 17–28.
- Loss, S., Siddique, K., Jettner, R., and Martin, L. (1998). Responses of faba bean (*Vicia faba* L.) to sowing rate in south-western Australia I. Seed yield and economic optimum plant density. *Aust. J. Agric. Res.* **49**, 989–997.
- Mason, I., and Fischer, R. A. (1986). Tillage practices and the growth and yield of wheat in southern New South Wales: Lockhart, in a 450 mm rainfall region. *Aust. J. Exp. Agric.* **26**, 457–468.
- Mayer, J. D., Lawn, R. J., and Byth, D. E. (1991). Adaptation of soybean [*Glycine max* (L.) Merrill] to the dry season of the tropics. II. Effects of genotype and environment on biomass and seed yield. *Aust. J. Agric. Res.* **42**, 517–530.
- McDonald, G. (2006). Effects of soil properties on variations in growth, grain yield and nutrient concentration of wheat and barley. *Aust. J. Exp. Agric.* **46**, 93–105.
- McKenzie, N., Jacquier, D., Isbell, R., and Brown, K., (Eds.) (2004). In “Australian Soils and Landscapes: An Illustrated Compendium”. CSIRO Publishing, Collingwood.
- McKirdy, S. J., Jones, R. A. C., and Nutter, F. W. (2002). Quantification of yield losses caused by barley yellow dwarf virus in wheat and oats. *Plant Dis.* **86**, 769–773.
- McMurray, L. (2005). Filed pea trials. “Crop Variety Evaluation 2004”, South Australian Research and Development Institute, Adelaide.
- McMurray, L., and Matic, R. (2003). Vetch trials. “Crop Variety Evaluation 2003”, South Australian Research and Development Institute, Adelaide.
- Mitchell, J. H., Fukai, S., and Cooper, M. (1996). Influence of phenology on grain yield variation among barley cultivars grown under terminal drought. *Aust. J. Agric. Res.* **47**, 757–774.
- Nanja Reddy, Y. A., Uma Shaanker, R., Prasad, T. G., and Udaya Kumar, M. (2003). Physiological approaches to improving harvest index and productivity in sunflower. *Helia* **26**, 81–90.

- Nicholls, N., Drosdowsky, W., and Lavery, B. (1997). Australian rainfall variability and change. *Weather* **52**, 66–71.
- Nix, H. A. (1975). The Australian climate and its effects on grain yield and quality. In “Australian field crops: Volume 1. Wheat and other temperate cereals” (A. Lazenby and E. Matheson, Eds.), pp. 183–226. Angus and Robertson, North Ryde.
- NVT. (2008). National Variety Trials Australian Crop Accreditation System (ACAS), Horsham (Online. Vol. 2008).
- Ockerby, S. E., and Fukai, S. (2001). The management of rice grown on raised beds with continuous furrow irrigation. *Field Crops Res.* **69**, 215–226.
- O’Connell, M., Connor, D., and O’Leary, G. (2002). Crop growth, yield and water use in long fallow and continuous cropping sequences in the Victorian mallee. *Aust. J. Exp. Agric.* **42**, 971–983.
- O’Leary, G. J., and Connor, D. (1997). Stubble retention and tillage in a semi-arid environment: 3. Response of wheat. *Field Crops Res.* **54**, 39–50.
- O’Mara, B. (undated). Chickpea Harvest and Seed Storage. Pulse Australia, Edgecliffe, NSW.
- Palta, J., and Fillery, I. (1995). N application increases pre-anthesis contribution of dry matter to grain yield in wheat grown on a duplex soil. *Aust. J. Agric. Res.* **46**, 507–518.
- Palta, J., and Ludwig, C. (1996). Pod set and seed yield as affected by cytokinin application and terminal drought in narrow-leaved lupin. *Aust. J. Agric. Res.* **48**, 81–90.
- Palta, J., and Ludwig, C. (1998). Yield response of narrow-leaved lupin plants to variations in pod number. *Aust. J. Agric. Res.* **49**, 63–68.
- Palta, J., and Ludwig, C. (2000). Elevated CO₂ during pod filling increased seed yield but not harvest index in indeterminate narrow-leaved lupin. *Aust. J. Agric. Res.* **51**, 279–286.
- Palta, J., and Plaut, Z. (1999). Yield and components of seed yield of indeterminate narrow-leaved lupin (*Lupinus angustifolius* L.) subjected to transient water deficit. *Aust. J. Agric. Res.* **50**, 1225–1232.
- Palta, J., Turner, N. C., French, R. J., and Buirchell, B. (2003). Selection for high lupin yield under terminal drought. “Agribusiness Crop Updates 2003”, Agriculture WA, Perth.
- Passioura, J. (1977). Grain yield, harvest index, and water use of wheat. *J. Aust. Inst. Agric. Sci.* **43**, 117–120.
- Passioura, J. (2006). Increasing crop productivity when water is scarce—From breeding to field management. *Agric. Water Manag.* **80**, 176–196.
- Pate, J. S., and Armstrong, E. L. (1996). Pea. In “Photoassimilate distribution in plants and crops” (E. Zamski and A. Schaffer, Eds.), pp. 625–642. Marcel Dekker Inc., New York.
- Pate, J. S., and Layzell, D. B. (1990). Energetics and costs of nitrogen assimilation. In “The Biochemistry of Plants” (B. Mifflin and P. Lea, Eds.), Vol. 16, pp. 1–42. Academic Press.
- Pate, J. S., Atkins, C. A., and Perry, M. W. (1980). Significance of photosynthate produced at different stages of growth as carbon source for fruit filling and seed reserve accumulation in *Lupinus angustifolius* L. *Aust. J. Plant Physiol.* **7**, 283–297.
- Penning de Vries, F. W. T., Brunsting, A. H. M., and Van Laar, H. H. (1974). Products, requirements and efficiency of biosynthesis a quantitative approach. *J. Theor. Biol.* **45**, 339–377.
- Penning de Vries, F. W., Van Laar, H., and Chardon, M. (1983). Bioenergetics of growth of seeds, fruits and storage organs. In “Potential Productivity of Field Crops Under Different Environments” (Anon, Ed.), pp. 37–59. International Rice Research Institute, Los Banos, Philippines.
- Perry, M. W., and D’Antuono, M. F. (1989). Yield improvement and associated characteristics of some Australian spring wheat cultivars introduced between 1860 and 1982. *Aust. J. Agric. Res.* **40**, 457–472.
- Perry, M. W., Delane, R. J., Tennant, D., and Hamblin, A. P. (1986). The growth and grain yield of lupins in relation to the soil water balance. “IV International Lupin Conference”,

- pp. 112–128. Western Australian Department of Agriculture: International Lupin Association, Geraldton.
- Pettersen, D. S., and Mackintosh, J. B. (1994). The chemical composition and nutritive value of Australian grain legumes. Grains Research and Development Corporation, Canberra.
- Prihar, S. S., and Stewart, B. A. (1990). Using upper-bound slope through origin to estimate genetic harvest index. *Agron. J.* **82**, 1160–1165.
- Prince, S., Haskett, J., Steininger, M., Strand, H., and Wright, R. (2001). Net primary production of the US mid-West croplands from agricultural harvest yield data. *Ecol. Appl.* **11**, 1194–1205.
- Pritchard, K., and Small, S. (1992). Soil and water management for irrigated crops on a red-brown earth soil. In “6th Australian Agronomy Conference” (D. Hutchinson and P. Vickery, Eds.), p. 251. Australian Society of Agronomy, Armidale.
- Reader, M. A., Dracup, M., and Atkins, C. A. (1997). Transient high temperatures during seed growth in narrow-leaved lupin (*Lupinus angustifolius* L.) II. Injurious high pod temperatures are likely in Western Australia. *Aust. J. Agric. Res.* **48**, 1179–1186.
- Richards, R., and Townley-Smith, T. (1987). Variation in leaf area development and its effect on water use, yield and harvest index of droughted wheat. *Aust. J. Agric. Res.* **38**, 983–992.
- Robertson, M. J., Holland, J. F., Cawley, S., Bambach, R., Cocks, B., and Watkinson, A. (2001). Phenology of canola cultivars in the northern region and implications for frost risk. “Science and technology: Delivering results for agriculture. Proc. 10th Australian Agronomy Conference”, Australian Society of Agronomy, Hobart.
- Robertson, M., Cawthray, S., Birch, C., Crawford, M., Dalgleish, N., and Hammer, G. (2003). Managing the risks of growing dryland maize in the northern region. In “5th Australian Maize Conference” (C. Birch and S. R. Wilson, Eds.), pp. 112–119. Maize Association of Australia, Toowoomba.
- Robertson, M. J., Holland, J. F., and Bambach, R. (2004). Response of canola and Indian mustard to sowing date in the grain belt of north-eastern Australia. *Aust. J. Exp. Agric.* **44**, 43–52.
- Robinson, D. (2004). Scaling the depths: Below-ground allocation in plants, forests and biomes. *Funct. Ecol.* **18**, 290–295.
- Rose, I., McCaffery, D., Lowien, J., and McGufficke, R. (2005). Soybeans: Inland northern NSW planting guide 2005–2006. Department of Primary Industries NSW Rep. No. 103.
- Sadras, V. O., and Connor, D. (1991). Physiological basis of the response of harvest index to the fraction of water transpired after anthesis: A simple model to estimate harvest index for determinate species. *Field Crops Res.* **26**, 227–239.
- Saxena, N. P., Natarajan, M., and Reddy, M. S. (1983). Chickpea, pigeon pea and groundnut. In “Potential productivity of field crops under different environments” (Anon, Ed.), pp. 281–305. International Rice Research Institute, Los Banos, Philippines.
- Schapaugh, W., and Wilcox, J. (1980). Relationships between harvest indices and other plant characteristics in soybeans. *Crop Sci.* **20**, 529–533.
- Schultz, J. E. (1995). Crop production in a rotation trial at Tarlee, South Australia. *Aust. J. Exp. Agric.* **35**, 865–876.
- Schwenke, G., Peoples, G., Turner, G., and Herridge, D. (1998). Does nitrogen fixation of commercial dryland chickpea and faba bean crops in north-west New South Wales maintain or enhance soil nitrogen? *Aust. J. Exp. Agric.* **38**, 61–70.
- Seymour, M. (1996). Response of narrow-leaved lupin (*Lupinus angustifolius* L.) to foliar application of growth regulators in Western Australia. *Aust. J. Exp. Agric.* **36**, 473–478.
- Sheehy, J. E., Peng, S., Dobermann, A., Mitchell, P. L., Ferrer, A., Yang, J., Zou, Y., Zhong, X., and Huang, J. (2004). Fantastic yields in the system of rice intensification: Fact or fallacy? *Field Crops Res.* **88**, 1–8.

- Shepherd, K. D., Cooper, P. J., Allan, A., Drennan, D., and Keatinge, D. (1987). Growth, water use and yield of barley in Mediterranean-type environments. *J. Agric. Sci. (Camb.)* **108**, 365–378.
- Shrestha, R., Turner, N. C., Siddique, K. H. M., Turner, D., and Speijers, J. (2006). A water deficit during pod development in lentils reduces flower and pod numbers but not seed size. *Aust. J. Agric. Res.* **57**, 427–438.
- Si, P., and Walton, G. (2004). Determinants of oil concentration and seed yield in canola and Indian mustard in the lower rainfall areas of Western Australia. *Aust. J. Agric. Res.* **55**, 367–377.
- Siddique, K. H. M., Belford, R. K., Perry, M. W., and Tennant, D. (1989). Growth, development and light interception of old and modern wheat cultivars in a Mediterranean-type environment. *Aust. J. Agric. Res.* **40**, 473–487.
- Siddique, K., Loss, S. P., Regan, K. L., and Pritchard, D. L. (1998). Adaptation of lentil (*Lens culinaris* Medik) to short season Mediterranean-type environments: Response to sowing rates. *Aust. J. Agric. Res.* **49**, 1057–1066.
- Siddique, K. M. H., Regan, K. L., Tennant, D., and Thomson, B. D. (2001). Water use and water use efficiency of cool season grain legumes in low rainfall Mediterranean-type environments. *Eur. J. Agron.* **15**, 267–280.
- Sinclair, T. (1993). Crop yield potential and fairy tales. In “International Crop Science 1” (D. Buxton, R. Shibles, R. Forsberg, B. Blad, K. Asay, G. Paulsen, and R. Wilson, Eds.), pp. 707–711. Crop Science Society America, Madison.
- Sinclair, T. (1994). Limits to crop yield? In “Physiology and determination of crop yield” (K. Boote, J. M. Bennett, T. Sinclair, and G. M. Paulsen, Eds.), pp. 509–532. ASA, CSSA, SSSA, Madison.
- Sinclair, T. (1998). Historical changes in harvest index and crop nitrogen accumulation. *Crop Sci.* **38**, 638–643.
- Sinclair, T., and De Wit, C. T. (1975). Photosynthate and nitrogen requirements for seed production by various crops. *Science* **189**, 565–567.
- Slafer, G., Araus, J., and Richards, R. (1999). Physiological traits that increase the yield potential of wheat. In “Wheat ecophysiology and yield determination” (E. Satorre and G. Slafer, Eds.), pp. 379–415. Food Products Press, New York.
- Smil, V. (1999). Crop Residues: Agriculture’s Largest Harvest. *Biosci.* **49**, 299–308.
- Snowball, R. (1986). Swathing field crops in the south-west. *J. Agric. West. Aust.* **27**, 84–88.
- Spenceley, J. (2005). Grain Sorghum. NSW Department Primary Industries, Sydney. Rep. No. P3.3.5.
- Stapper, M., and Fischer, R. A. (1990a). Genotype, sowing date and plant spacing influence on high-yielding irrigated wheat in southern New South Wales. I. Phasic development, canopy growth and spike production. *Aust. J. Agric. Res.* **41**, 997–1019.
- Stapper, M., and Fischer, R. A. (1990b). Genotype, sowing date and plant spacing influence on high-yielding irrigated wheat in southern New South Wales. II. Growth, yield and nitrogen use. *Aust. J. Agric. Res.* **41**, 1021–1041.
- Stapper, M., and Fischer, R. A. (1990c). Genotype, sowing date and plant spacing influence on high-yielding irrigated wheat in southern New South Wales. III. Potential yields and optimum flowering dates. *Aust. J. Agric. Res.* **41**, 1041–1056.
- Stone, P., and Nicholas, M. (1995). A survey of the effects of high temperature during grain filling on yield and quality of 75 wheat cultivars. *Aust. J. Agric. Res.* **46**, 475–492.
- ten Berge, H. F. M., Thiyagarajan, T. M., Shi, Q., Wopereis, M. C. S., Drenth, H., and Jansen, M. J. W. (1997). Numerical optimization of nitrogen application to rice. Part I. Description of MANAGE-N. *Field Crops Res.* **51**, 29.
- Thomas, G. (2005). Grain yield and protein content for cropping systems at Dunkerry south trial 1996–2001. CSIRO, Toowoomba. Vol. 2006.

- Thomas, Fukai, S. (1995). Growth and yield response of barley and chickpea to water stress under three environments in southeast Queensland. I. Light interception, crop growth and grain yield. *Aust. J. Agric. Res.* **46**, 17–33.
- Thompson, J. A., and Heenan, D. P. (1989). The time of sowing of irrigated grain sorghum in the Murrumbidgee Valley. In “5th Australian Agronomy Conference” (M. W. Perry, Ed.), p. 509. Australian Society Agronomy, Perth.
- Thomas, Robertson, M. J., Fukai, S., and Peoples, M. B. (2004). The effect of timing and severity of water deficit on growth, development, yield accumulation and nitrogen fixation of mungbean. *Field Crops Res.* **86**, 67.
- Turner, N. (1997). Further progress in crop water relations. *Adv. Agron.* **758**, 293.
- Turner, N. C., Leport, L., French, R. J., Thomson, B. D., and Siddique, K. H. M. (1996). Adaptation of cool-season pulses to water-limited environments. In “8th Australian Agronomy Conference” (M. Asghar, Ed.), pp. 554–557. Australian Society of Agronomy, Toowoomba.
- Turner, N., Wright, G., and Siddique, K. H. M. (1999). Adaptation of grain legumes (pulses) to water-limited environments. *Adv. Agron.* **71**, 193–229.
- Unkovich, M. J. (1991). The potential of the $\delta^{15}\text{N}$ natural abundance method for estimating the contribution of fixed nitrogen to the nitrogen economy of broadacre lupins (*Lupinus angustifolius*), M.Sc. Thesis, The University of Western Australia.
- Unkovich, M. J., Pate, J. S., and Hamblin, M. J. (1994). The nitrogen economy of broadacre lupin in southwest Australia. *Aust. J. Agric. Res.* **45**, 149–164.
- Unkovich, M. J., Baldock, J., and Forbes, M. (2006). Australian Crop Yields and Harvest Indices (Microsoft Access Database). CSIRO Land and Water, Adelaide.
- Unkovich, M., Baldock, J., and Marvanek, S. (2009). Which broadacre crops and pastures should be included in a carbon accounting system for Australia’s cropping zone? *Crop Pasture Sci.* **60**, 1–10.
- van Herwaarden, A., and Richards, R. (2002). Water-soluble carbohydrate accumulation in stems is related to breeding progress in Australian wheats. In “Plant Breeding for the 11th Millennium; 12th Australian Plant Breeding Conference” (J. McComb, Ed.), pp. 878–882. Australasian Plant Breeding Conference, Perth, Vol. Compact Disc.
- van Herwaarden, A., Farquhar, G., Angus, J., Richards, R. A., and Howe, G. (1998a). ‘Haying-off’, the negative grain yield response of dryland wheat to nitrogen fertiliser. I. Biomass, grain yield and water use. *Aust. J. Agric. Res.* **49**, 1067–1081.
- van Herwaarden, A., Richards, R. A., Angus, J., and Farquhar, G. (1998b). Haying-off, the negative grain yield response of dryland wheat to nitrogen fertiliser. II Carbohydrate and protein dynamics. *Aust. J. Agric. Res.* **49**, 1083–1093.
- Vleeshouwers, L. M., and Verhagen, A. (2002). Carbon emission and sequestration by agricultural land use: A model study for Europe. *Glob. Change Biol.* **8**, 519–530.
- Wahbi, A., and Gregory, P. J. (1989). Genotypic differences in root and shoot growth of barley (*Hordeum vulgare*) II. Field studies of growth and water use of crops grown in northern Syria. *Aust. J. Exp. Agric.* **25**, 389–399.
- Ward, P. R., Dunin, F. X., and Micin, S. F. (2002). Water use and root growth by annual and perennial pastures and subsequent crops in a phase rotation. *Agric. Water Manag.* **53**, 83–97.
- Wheeler, R., and McMurray, L., (Eds.) (2000). In “South Australian Field Crop Evaluation Program Annual Report 1999”. South Australian Research and Development Institute, Adelaide.
- Whitehead, S. J., Summerfield, R. J., Muehlbauer, F. J., Coyne, C. J., Ellis, R. H., and Wheeler, T. R. (2000). Crop improvement and the accumulation and partitioning of biomass and nitrogen in lentil. *Crop Sci.* **40**, 110–120.
- Williams, R. L., and Angus, J. (1994). Deep floodwater protects high-nitrogen rice crops from low-temperature damage. *Aust. J. Exp. Agric.* **34**, 927–932.

- Wright, P., Morgan, J., and Cass, A. (1992). Effects of water stress on yield and the components of yield in canola and Indian mustard. "6th Australian Agronomy Conference", p. 617. Australian Society of Agronomy, Armidale.
- Zhang, X., and Evans, P. M. (2004). Grain yield production in relation to plant growth of wheat and canola following clover pastures in southern Victoria. *Aust. J. Exp. Agric.* **44**, 1003–1012.
- Zhang, J., and Yang, J. (2004). Crop yield and water use efficiency: A case study in rice. In "Water Use Efficiency in Plant Biology" (M. A. Bacon, Ed.). Blackwell Publishing/CRC Press, Oxford.
- Zubaidi, A., McDonald, G. K., and Hollamby, G. J. (1999). Shoot growth, root growth and grain yield of bread and durum wheat in South Australia. *Aust. J. Exp. Agric.* **39**, 709–720.
- Zwer, P., and Hoppo, H. (2004). Oat trials. In "South Australian Field Crop Evaluation Program; Post Harvest Report 2003/2004" (R. Wheeler and L. McMurray, Eds.), South Australian Research and Development Institute, Adelaide.